Too much and too little in Limbu conjugation

Syncretism, uninflectedness and dependent multiple exponence in Information-based Morphology

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Mémoire de M2 en Linguistique Théorique et Expérimentale

Université Paris Cité

16 June 2023

Linguistics
GRADUATE SCHOOL

Laboratoire de linguistique formelle

Abstract

I examine the conjugation system of Limbu (van Driem 1987), which features person–number agreement in all A, S and P roles, and I provide a full analysis of the indicative affirmative paradigm in Information-based Morphology (IbM), a formal theory of inflection couched in a logic of typed feature structures (Crysmann & Bonami 2016). Despite clear regularities, this system showcases several kinds of disruptions in the canonical one-to-one morphological mapping between form and function. One finds systematic number syncretism in several areas of the paradigm, as well as seemingly unmotivated ("morphomic") syncretism for certain markers, which escape simple accounts in terms of natural descriptions or default–override ("Pāṇinian") dynamics. However, a slightly broader view of the principles behind the organization of morphological splits can easily account for these patterns, including a partially motivated kind of split which I call "pseudo-Pāṇinian". I show this using the type abstraction and inheritance mechanisms in IbM, yielding an insightful account of the phenomena at hand, which questions the sharp divide between motivated and unmotivated splits. In particular, my analysis uses a distinction between general rule description and specific rule instances to capture these partly motivated splits, but also the systematic uninflectedness (too little marking) and dependent–carrier multiple exponence (too much marking) found in Limbu. This work therefore advances understanding of morphological splits thanks to close, logically informed inspection of existing data, and showcases on an intricate dataset the interest of sophisticated knowledge-representation devices for inflectional morphology.

Acknowledgments

Foremost, I wish to thank my supervisor, Berthold Crysmann, for guiding me enthusiastically throughout this year-long project. He is a clear-sighted and knowledgeable linguist, and provided many enlightening insights on formal morphology. But above all, I am truly grateful to him for his unwavering availability, patience and support across time and space, which made this experience quite unique. His advice and high standards in rhetoric, which I still fall short of even though I shouldn't by now, played a significant role in refining this thesis.

My deep thanks must also go to everyone, from mentors and role models to regulars and passers-by, who made linguistics feel like a welcoming place. Among others, Olivier Bonami for his supervision and advice; Aimée Lahaussois for her instructive, meticulous work, which was the original impetus for this two-year enterprise on Kiranti conjugation; Salvador Mascarenhas for distracting me from what's important (or is it, really?). For their advice and generously offering their time and thoughts: Karen De Clercq, Erich Round, Maria Copot, Mark Gorenstein.

Company is a great part of the journey, which is why I should name, in rough association-of-ideas order: Jeanne R., Alexandre R. and the karate crew; Niko, Matéo, Lucas, Bru, Elies, Aurore, Charles, Victor, Elodie; Lulu, Baba; Anissa, Ioanna, Thomas, Sungwoo, Asher, Judith, and the fluctuating but irreplaceable Luiz Arthur. And because because *on change pas une équipe qui gagne*: «les Huns» (Flavien, Martin, Robin; and Nono!). Finally, Nore and Noah, and my family, large and small. Thank you all.

Contents

Chapter 1

Introduction

This Master's thesis examines conjugation in Limbu (van Driem 1987), a Kiranti language spoken in eastern Nepal by 180,000 people. Similar to other Kiranti languages, such as Athpare (Ebert 1997a) or Camling (Ebert 1997b), Limbu verbs inflect for their core arguments, two in number for the transitive case. The conjugation systems of these distantly related languages are similar in that they are quite transparent, to the extent that they separate marking for subject and object, and furthermore, recognize different markers for person and number. Additionally, they all feature a handful of markers which bear striking similarities. They differ significantly, however, in what strategies they employ for specific cells.

In Limbu, the transparency of conjugation was noted by van Driem (1987: p. 69), who even characterizes the system as "efficient". Nevertheless, in line with other languages of the family, this is obscured by disruptions of different kinds, precisely like those that Matthews (1972) considered characteristic of inflectional systems. These come in the form of fusional exponence with portmanteau markers, systematic syncretism and markers which cannot be neatly described in terms of inflectional features. More specifically, on the one hand, the number of morphs to express a certain function can be unexpected; on the other, what marking appears can be hard to interpret precisely.

The first issue, in the function-to-form mapping, is that the amount of marking is variable. In cells with dependent multiple exponence (Harris 2017), one finds several identical markers with apparently identical functions, which is more than needed. By contrast, some properties may not be formally marked at all. Unlike the meaningful lack of marking in zero exponence, Limbu finds cells which are entirely uninflected for number: no markers appear even though they could; this is fewer than needed. While multiple exponence instantiates a $1 : n$ relation between function and form, and zero exponence a $1 : 0$ relation, uninflectedness can be understood as a "$0 : 0$" relation, where no function is even expressable anymore.

In fact, because certain functions cannot be expressed, the extent to which number is expressed for transitive verbs can be said to range roughly from uninflectedness for both participants (0 : 0) to full contrast for both (2 : 2). In between, one finds inflectedness for one participant only (1 : 1), and inflectedness with partial collapse of distinctions for one of the two ($2 : 1\frac{1}{2}$).

More precisely, collapse of number distinctions is one example where the interpretation of the markings that do appear is not straightforward. This is a qualitative shortcoming in the form-to-function mapping. Then, certain dual markers can take over plural cells, and conversely, certain plural markers can take over dual cells, in other parts of the paradigm. This gives rise to splits that are not fully motivated, which I will call "pseudo-Pāṇinian". Among non-natural splits, so-called Pāṇinian splits enjoy a special role, where a default is overridden in specific cells, yielding an unbalanced pattern that is nevertheless clearly understood in terms of a principle of knowledge use. With Limbu pseudo-Pāṇinian splits, the override itself is generalized such that we cannot speak of a more specific condition in the strict sense. In order to understand the nature of these splits, it is therefore useful to explore how they relate exactly to standard Pāṇinian competition.

The goal of this work is to describe these phenomena in Limbu, and to give an analysis that will account for them all. The theory of inflectional morphology I will use is Information-based Morphology (henceforth IbM; Crysmann & Bonami 2016, 2017).

The structure of this dissertation is laid out as follows. Chapter 2 contains a general discussion of types of morphological splits, with a view to comparing certain distributions in Limbu to morphomic patterns. Chapter 3 starts off with a tour of the Limbu system, outlining its general properties, and contrasting its transparency with a number of deviations. It gets to the heart of the matter by describing in more detail several pseudo-Pāṇinian splits, and multiple exponence. Section 3.3 discusses implications of patterns found in Limbu for morphological theories, and sets the stage for the analysis in Chapter 4. There, I develop a formal analysis of the conjugation system using IbM, whose formal specificities are presented in Section 4.1. I go on to explain how to analyze pseudo-Pāṇinian splits with a formal, subsumption-based version of Pāṇini's principle, as embraced by IbM, and then show how to reduce the morphomic splits to general patterns of competition and conditioning between non-singular markers in Limbu. Finally, I address how dependent copying and uninflectedness can be rendered in IbM by recognizing two orthogonal dimensions, one generally catering to the shape of markers, and the other roughly catering to their distributions. Chapter 5 concludes on pseudo-Pāṇinian splits and discusses the interaction of seemingly morphomic patterns with formal morphology, highlighting the strong points of IbM for the cluster of phenomena found in Limbu. It closes by mentioning closely related problems in other Kiranti languages.

Chapter 2

Morphological splits

Morphological inflection and agreement can be understood as the word-level reflection of syntax and semantics. Syntax and semantics group together certain types of context, thus providing dimensions along which words can vary. This shapes morphological paradigms according to inflectional features: inflected forms will then vary across cells. In the limiting case, every lexeme in the language can distinguish all cells in its paradigm; nevertheless, we often observe identity of forms between different cells, which is called syncretism.

Where distinctions are maintained, the variation can be straightforward, but frequently, several groups of formal patterns can be discerned in the paradigm: this is an inflectional split. The split can occur at different syntagmatic positions in the word: the stem, the affixes, or some suprasegmental feature. In this way, we can break up word-level splits into the conjunction of simpler, lower-level splits (Feist et al. 2021).

Several kinds of splits have been outlined by morphologists (Corbett 2015). German noun morphology has an exceptionally low functional load, as it features pervasive whole-form syncretism, making it a good ground for illustrating the types of splits. First, a **natural split** will only be sensitive to a few featural dimensions in the paradigm. Consider Table 2.1a: the number dimension is relevant to inflecting OMA 'granny' because the singular cells behave differently from the plural cells. These are so-called "natural" groupings, that follow the divides provided by the layout of the paradigm into featural dimensions. Thus, inflection is sensitive to, and split along, number. On the other hand, it fails to mark case, because natural classes such as "the singular cells" are indiscriminate of case.

On the contrary, some splits do not seem to run along particular featural divides. Instead, they contrast a wide-denotation default with a narrow-denotation exception, or override. They are called **Pāṇinian splits** because the more specific pattern takes precedence over the more general one; this is Pāṇini's principle (or the Elsewhere Condition: Kiparsky 1985). Usually, when one part of the split stands out as lacking some material, it tends to be the default; as in the case of WAGEN 'vehicle' in Table 2.1b where all cells

'granny'	SINGULAR	PLURAL
NOM	Oma	Oma-s
GEN	Oma	Oma-s
DAT	Oma	Oma-s
ACC	Oma	Oma-s

(a) Natural split

'vehicle'	SINGULAR	PLURAL
NOM	Wagen	Wagen
GEN	Wagen-s	Wagen
DAT	Wagen	Wagen
ACC	Wagen	Wagen

(b) Pāṇinian split

'person'	SINGULAR	PLURAL
NOM	Mensch	Mensch-en
GEN	Mensch-en	Mensch-en
DAT	Mensch-en	Mensch-en
ACC	Mensch-en	Mensch-en

(c) Pāṇinian split with zero override

'car'	SINGULAR	PLURAL
NOM	Auto	Auto-s
GEN	Auto-s	Auto-s
DAT	Auto	Auto-s
ACC	Auto	Auto-s

(d) Combination of natural and Pāṇinian split

Table 2.1: Motivated splits (from Crysmann 2021b: p. 957)

except GEN.SG contrast by their absence of an *-s*. Albeit more rarely, the converse is also found, as with MENSCH in Table 2.1c: even though *Mensch* is the base form with a zero exponent, it appears as the accidental exception within the paradigm, and when compared to other lexemes like invariant BREZEN 'pretzel' (Crysmann 2021b: p. 956).

Thirdly, some cases do not seem to fall into either category: the distribution is not wholly motivated by maintaining distinctions in one feature, nor by a single, featurally coherent override. They are called **morphomic splits**. Consider the two Romance examples in Table 2.2, studied extensively by Martin Maiden. Stem selection and function do not seem to neatly align; even though some generalizations are possible, for subjunctive or singular cells, respectively. One could conceive a combination of a natural and Pāṇinian split could be conceived for these two patterns, as in the case of German AUTO 'car' in Table 2.1d. However, their identity of form seems to persist across time, so much so that it stands out as a self-reinforcing, first-class morphological object (Maiden 2005). Contrast this with the German case, where genitive *-s* (Table 2.1b) and plural *-s* (Table 2.1a) exist independently of their coincidental interaction in the AUTO paradigm.

The power of generalization seems to be entirely lost, however, in exotic gender systems such as those of New-Guinean Burmeso (Corbett 2015), or of Caucasian Batsbi in Table 2.2c (Harris 2009). The recurrent use of certain forms is certainly a striking feature of the Batsbi paradigm, but syncretism does not seem to follow any justification in keeping the different genders apart. Every logically possible singular–plural paradigm with *y* and *d* is found for some gender in the language, for instance. Even the tendency

'say'	PRS.IND	PRS.SBJV
1SG	dig-o	dig-a
2SG	diz-es	dig-as
3SG	diz	dig-a
1PL	diz-emos	dig-amos
2PL	diz-eis	dig-ais
3PL	diz-em	dig-an

(a) The L-pattern morphome (Portuguese; Maiden 2005: p. 147)

'go'	PRS.IND
1SG	va-do
2SG	va-i
3SG	va
1PL	gi-mo
2PL	gi-te
3PL	va-nno

(b) The N-pattern morphome (Old Tuscan; Maiden 2005: p. 154)

Gender	SG	PL
I	v	b
II	y	d
III	y	y
V	d	d
VI	b	d
(IV)	b	b
(VII)	b	y
(VIII)	d	y

(c) Batsbi (Tsova-Tush) gender markers

Table 2.2: Several morphomic splits

for more natural semantic classes in the plural, pointed out by Baerman, Brown & Corbett (2005: p. 86) does not seem very active in such cases.

Morphological theories usually have devices for both the first types of splits, but the third can be more elusive. In particular, underspecification can easily account for natural splits because it allows to ignore as many featural divides as needed for a particular split. Some version of Pāṇini's principle with the ability to underspecify markers can readily take care of Pāṇinian splits. These devices only work when there is some featural motivation to the splits, however. This raises the question of how to capture identity of form in outlier cases bordering on morphomes.

Consider the dual number marker **-ci* in Kiranti languages (van Driem 1990: p. 47): it usually serves to mark dual number in verbs. Table 2.3 witnesses this for the reflex *-si* in the transitive conjugation of Limbu, where the dual cells are spread out in rows (for subjects) and columns (for objects), drawing a striped pattern. This clean, natural pattern is disrupted in many corners of the paradigm, however. Even though the lower-left "inverse" zone features dual agents, which should give rise to horizontal stripes, no such stripes are found as *-si* is not available. In other corners such as the right-hand 3PL column, *-si* is generalized and serves to express general-purpose non-singular number. The multidimensional nature of the system obfuscates the patterns that can be found, but the motivations for including certain cells while excluding others do not seem quite as elusive as in the cases of morphomes above.

In the spirit of Baerman, Brown & Corbett (2005)'s study on syncretism, I will endeavor to account for these phenomena in Limbu, in the framework of Information-based Morphology (Crysmann & Bonami 2016, 2017).

↓A \ P→	1SG	1DE	1PE	1DI	1PI	2SG	2DU	2PL	3SG	3DU	3PL
1SG						—	—	—	—	—	—
1DE						—	—	—	—	—	—
1PE						—	—	—	—	—	—
1DI									—	—	—
1PI									—	—	—
2SG	—	—	—						—	—	—
2DU	—	—	—						—	—	—
2PL	—	—	—						—	—	—
3SG	—	—	—	—	—	—	—	—	—	—	—
3DU	—	—	—	—	—	—	—	—	—	—	—
3PL	—	—	—	—	—	—	—	—	—	—	—
S→	—	—	—	—	—	—	—	—	—	—	—

Table 2.3: Distribution of *-si* in Limbu. Darker cells signal the cooccurrence of two *-si* markers.

Chapter 3

The Limbu conjugation system

The multidimensional Limbu conjugation system contains several unusual splits and distributions. Here, I describe the splits in the context of the marking strategies of the system, outline their specific problems, and connect them to the morphome debate. The section ends on the phenomenon of morphotactic dependence and the account of it that was provided by Stump (2022).

3.1 Description of the system

This section first provides a general overview of the Limbu simplex verb and introduces the more transparent aspects of participant marking in Section 3.1.2. In Section 3.1.3, I shall discuss some interesting deviations form a canonical agglutinative system; including fusion, zero exponence, multiple exponence, and syncretism.

3.1.1 General overview

For the purposes of this study, I am only concerned with what van Driem terms "simplex" verbs, i.e. synthetic forms which inflect for person and tense but not aspect or mood (van Driem 1987: p. xxvii). Aspect and mood can be encoded with more peripheral material, affixes or periphrastic constructions, added to simplex forms (van Driem 1987: ch. 5–7). Two inflectional dimensions are relevant for simplex forms in all verbs: tense (preterite or non-preterite) and polarity (affirmative or negative). Additionally, depending on transitivity, one or two arguments must be encoded on the verb: this adds one or two macro-dimensions to each paradigm. Within each macro-dimension, the verb simultaneously inflects for person (four values, with clusivity) and number (three values: singular, dual, plural) of a given argument.

The place that an argument takes in the core of a clause is modeled using a macrorole, or simply "role". For an intransitive verbs, the only argument it subcategorizes is labeled with an "S" role, sometimes "(intransitive) subject" in the text. Following Comrie (1978), in transitive verbs, the more agentive of the two arguments is called "A" (sometimes, abusively, "agent" or "transitive subject" in the text) whereas the less agentive has a so-called "P" role (sometimes "patient" or "object"). Every ditransitive verb patterns like a transitive in that it indexes two of its arguments: the more agent-like automatically, and usually the Recipient or beneficiary instead of the Theme for animacy reasons (van Driem 1987: 272f; notations from Siewierska 2004: p. 57); indexation of T does occur and is generally lexically determined (van Driem 1987: p. 275). Thus, no special label is necessary, and I follow van Driem in considering them members of the morphological class of transitive verbs. Like Lahaussois (2020: p. 40), I use "scenario" to mean a specific, simultaneous combination of person–number for all arguments within a given argument structure (possibly just S, or else both A and P). I will use a compact notation to represent such scenarios: σ>τ for a scenario in which the A has features σ and the P has features τ. Sometimes I reverse the arrow to mean the opposite relation: τ<σ. [1]

Both for intransitive and transitive verbs, the whole system largely behaves agglutinatively and suffixally; though some prefixes are recurrent, unlike such Kiranti languages as Thulung which only mark polarity and some ideophones prefixally (Lahaussois 2020: p. 41). Therefore, the aforementioned paradigm dimensions are generally independent, with a 1 : 1 correspondence: the marker for each property and the property for each marker are easily identified. In Limbu participant marking, the independence of different dimensions goes to such an extent that person and number often have separate exponents, for a single participant.

This broad-stroke description does not prevent boundary effects in the system, which I will get to shortly. On the one hand, there is some fairly tame fusional participant and tense marking, which outcompetes more discrete marking. Other non-canonical forms of exponence (zero, extended, overlapping) are used, for isolated parts of the system. On the other hand, a more difficult puzzle in the system is the relative uninflectedness of certain cells (sometimes closer to "canonical syncretism" in the terminology of Baerman, Brown & Corbett 2005: p. 27): they feature less material than they could in principle, even collapsing some features entirely. These phenomena are quite specific, however, and do not disrupt the robust form-to-function association that one can draw for most markers.

[1] In double entry tables, I may vary the features of a given participant in one axis, and in the other axis the role of this participant (and possibly the other participant it interacts with). Then, I write "σ_1" in the first axis and, redundantly, "A>σ_2/P<σ_2" to emphasize the role that participant with features σ_1 takes on in a specific cell. At the intersection of a row and a column, the scenario will then be "σ_1>σ_2/σ_1<σ_2".

−5..−3 pf1	−2..−1 pf2-3	0 (root)	+1 sf1	+2 sf2	+3..+7 sf3-10	+8 sf11	+9 sf12
kε-: 2	***n-***: NEG		*-siŋ*: REFL	***-ε***: PRET	***-si***: DU	*-ge*: 1EXCL	***-n***: NEG
mε-: 3PL	(*mε-*: NEG)		*-nε*: 1>2	∅: NPRET	*-i*: PL		(*-nεn*: NEG)
a-: 1INCL	∅: AFF				∅: SG		
					-u: 3P		

Table 3.1: Morphological template for Limbu conjugation. Exponents used in (1) are boldfaced. Negative exponence is extended in slots −2..−1 and +9. Allomorphic variants are in parentheses.

3.1.2 A more involved dive into verbal template and paradigms

Properties of the intransitive paradigm

Having laid down these remarks about the general nature of the system, a preview is in order. I will use the intransitive paradigm as a stepping stone into the transitive one. The division of morphological labor, and the resulting transparency, is showcased by forms such as (1).

(1) kε-n- nu:ks- ε-tchi-n (**boldfaced** in Table 3.3)
2-NEG- return- PRET-DU-NEG [2DUS]
'You[du] did not return.' (van Driem 1987: p. 375)

There, we can already identify the usual marker of preterite tense, *-ε*, and the extended exponence of negation. These, along with person (*kε-*) and number (*-tchi*, allomorph of *-si*) exponence, fit into an agglutinative template, a simplified version of which is shown in Table 3.1. I use my own labeling for the position classes, in coherence with the rest of the text and especially Sections 4.2–4.5. Differences with van Driem (1987)'s or Stump (2022, 2017: 93, 423ff)'s analyses are not relevant here as they are masked within the slot intervals.

On the whole, however, it is worthwhile noticing an unusual separation for fusional Indo-European languages, or even agglutinative Swahili: person and number occupy different slots, with decorrelated markers. Thus, *kε-* is used for other second-person forms; likewise, *-si* is used for other dual forms. In languages with cumulative markers, the person–number "macro-dimension", exemplified when varying just one argument of any verb (the S of an intransitive or the A/P of a transitive), only has one paradigmatic series of exponents, and thus behaves like one inflectional dimension instead of two independent ones.

Removing negative and preterite marking from (1) allows one to recover the affirmative non-preterite form of the verb (barring predictable stem allomorphy): *kε-nu:ŋ-si*, which therefore can be said to feature zero exponence of affirmative and non-

Type of marker	Form	Content	Condition
Person (−5..−2)	∅	3	
	kɛ-	2	
	a-	1INCL	
	-ge	1EXCL	
Number (+3..+7)	∅	SG	
	-si	DU	
	-i	PL	1EXCL∨2
	∅	PL	1INCL
Cumulative (−2/+3)	*mɛ-*	3PL	
	-mʔna	1PE.PRET	
	-ʔɛ	1SG.NPRET	
	-aŋ	1SG.PRET	

Table 3.2: Person markers in the Limbu intransitive paradigm. Conditioning is determined on the basis of Pure Sensitivity (Carstairs 1987).

S →	1SG	1DE	1PE	1DI	1PI	2SG	2DU	2PL	3SG	3DU	3PL
	-ʔɛ / -aŋ	-si-ge	-i-ge / -mʔna	a- -si	a-	kɛ-	**kɛ- -si**	kɛ- -i	∅	-si	mɛ-

Table 3.3: Simplified intransitive paradigm

preterite (shown in Table 3.1, as van Driem 1987: p. 75). Other intransitive forms of the paradigm feature different person and number markers than *kɛ-* 2 and *-si* DU: the contrasting exponents are listed in Table 3.2: as can be seen there, roughly one marker is found for each feature value. Several fusional markers are already visible, restricted to first person and third person plural. The exponents in Table 3.2 (and their slots as shown in Table 3.1), as well as run-of-the-mill Pāṇinian competition are sufficient to inflect any intransitive verb. This can be checked in Table 3.3, which shows the intransitive Limbu paradigm. For instance, the first person dual exclusive form of any verb is expressed with the suffixes *-si-ge* -DU-1EXCL. As the only exception, the first person plural inclusive form only has the prefix *a-* for 1INCL and no overt plural expression. The subdivisions in the 1SGS and 1PES cells of Table 3.3 represent significant tense and polarity allomorphy: I divide all cells with such phenomena into four, with non-preterite above, preterite below, affirmative on the left and negative on the right.

Now that the intransitive picture is clear, let us turn to transitive scenarios. As explained in the previous section, transitive verbs have two macro-dimensions for participant marking instead of one, for intransitives. Therefore, the one-dimensional Table 3.3, where tense and polarity are mostly abstracted away, will now become a two-dimensional one. By convention, this will be a double-entry table, with agent fixed in rows and patient fixed in columns. The layout has been common in Kiranti descriptions since the 1970s (Lahaussois 2019).

The result of compiling transitive person material, for all scenarios, is shown in Table 3.4. This table warrants a few remarks. The intransitive forms are recalled (in greater detail) as the last row; they could be displayed as the last column, too, as does Ebert (1997a,b) in her Athpare and Camling grammars. Like her, I use arrows to represent syncretism between adjacent cells, without necessarily implying any directionality. As is usual in other Kiranti languages and elsewhere,[3] transitives in general have the restriction that they cannot index two arguments which partially corefer: this leaves the "reflexive-like" scenarios of each paradigm empty, along the diagonal.Instead, reflexive and reciprocal configurations are conveyed using "derived" reflexivized verbs which are therefore intransitive and mark person identically to other intransitives.[4] Full paradigms with forms for all values of tense and polarity can be found in appendix C.

Properties of the transitive paradigm

Much of what was said about intransitives still holds for the transitive conjugation. Tense and polarity marking obey the same rules. Marking falls into the template of Table 3.1, though expanded to take into account the longer suffix chains. The interpretation of many markers extends seamlessly into the transitive paradigm: this exemplifies the great agglutinativity and context-independence of marking.

As a first testimony of this coherence, notice that the 3SGA row is nearly identical to the bottom intransitive row: thus, markers that served to mark the S participant in fact receive an interpretation which is largely role-independent. In fact, this is an instance of heteroclisis: one can say that the intransitive paradigm reuses the corresponding 3SG>P forms in nearly all cells. On the other hand, the rightmost cell of the intransitive row differs from the corresponding 3SG>P cell: it instead uses the material from the A>3SG cell (bottom of 3SGP column), removing the predictable *-u* 3P marker. The intransitive paradigm sits in the middle, sometime patterning like a "patient paradigm" and

[2]Cells with allomorphy conditioned by tense or polarity are divided up into 2-by-2 subtables, with non-past at the top, past at the bottom, affirmative on the left and negative on the right. Regular tense and polarity marking has been omitted from the paradigm in the interest of readability (largely reproducing the non-past affirmative paradigm).

[3]In other languages that index two arguments on their verbs: Papuan Yimas uses a valency-decreasing reciprocal affix and may use periphrasis or formally intransitive affixes on a transitive root for reflexives (Foley 1991: 217, 284ff), Australian Iwaidja (Evans 2000: pp. 111–113).

[4]This is my own observation, though it is present implicitly in van Driem (1987): he calls "subject" any argument of an intransitive or reflexive verb, and intransitive and reflexive conjugations are mentioned together throughout chapter 4. One exception is that the reflexive suffix *-siŋ* itself has an allomorphic variant *-nɛ* when inflecting for dual S (van Driem 1987: p. 86).

↓A \ P →	1SG	1DE	1PE	1DI	1PI	2SG	2DU	2PL	3SG	3DU	3PL
1SG						-nɛ	-nɛ-tchi-ŋ	-n-i-ŋ	-u-ŋ | -ʔɛ / -paŋ[a]	-u-ŋ-si-ŋ | -ʔɛ-n-chi-n[b] / -paŋ-si-ŋ[a]	→
1DE						←	-nɛ-tchi-ge	→	-s-u-ge	-s-u-si-ge	→
1PE						↙	↓	↘	-u-m-be / -mʔna[a]	-u-m-si-m-be / -mʔna-si[a]	→
1DI									a- -s-u	a- -s-u-si	→
1PI									a- -u-m	a- -u-m-si-m	→
2SG	kɛ- -ʔɛ / kɛ- -aŋ	↑	↗						kɛ- -u	kɛ- -u-si	→
2DU	←	a-gɛ-	→						kɛ- -s-u	kɛ- -s-u-si	→
2PL	↙	↓	↘						kɛ- -u-m	kɛ- -u-m-si-m	→
3SG	-ʔɛ / -aŋ	-si-ge	-i-ge	a- -si	a-	kɛ-	kɛ- -si	kɛ- -i	-u	-u-si	→
3DU	↑	↑	↑	↑	↑	↑	↑	↑	-s-u	-s-u-si	→
3PL	mɛ- -ʔɛ / mɛ- -aŋ	mɛ- -si-ge	mɛ- -i-ge	a-m- -si	a-m-	kɛ-m-	kɛ-m- -si	kɛ-m- -i	mɛ- -u	mɛ- -u-si	→
S →	-ʔɛ / -aŋ | -paŋ[a], -aŋ	-si-ge	-i-ge / -mʔna[a]	a- -si	a-	kɛ-	kɛ- -si	kɛ- -i	∅	-si	mɛ-

[a] Does not take the expected negative suffix *-n(ɛn)*. Instead, takes an additional prefix *n-* between usual negative *mɛ-* and the root (van Driem 1987: p. 104).

[b] Obligatorily features two negative suffixes *-n* instead of one. No additional suffix is needed; the usual *mɛ-* prefix is still required.

Table 3.4: Limbu person marking[2] (based on conjugation lists in van Driem 1987: pp. 368–374)

sometimes like an agent paradigm. This gives a first taste of the different patterns of different markers.

Another generalization that remains true is that different dimensions are often kept separate, even when two features are involved for each participant. Consider the two forms in 1DI>3PL and 2PL>3PL. The first, *a- -s-u-si* 1INCL- -DUA-3P-NSGP, employs four markers for four properties.

The generality of markers sometimes makes them pervasive tools in the Limbu speaker's toolbox. For instance, *kɛ-* 2 (and its allomorph *gɛ-*) is expected in all scenarios involving a second-person, and only those: that is, any cell within the three second-person rows or the three second-person columns, and no other. This generalization almost holds: *kɛ-* is not found in the upper-center part of the table. The reason for this is a more specific fusional suffix *-nɛ* which will be explained in the next section. Apart from this, *kɛ-* still works together with number expression just like it did in the intransitive scenario (see *kɛ- -si* in 3SG>2DU); but it now cooccurs happily with old and new affixes used to inflect for the other participant: *kɛ- -ʔɛ* in the 2SG>1SG.NPRET, *kɛ-m-* in 3PL>2SG (*m-* is an allomorph of *mɛ-* 3PL). The same hold of the exclusive marker *-ge* (allomorph *-be*): we expect it in all cells of the two exclusive rows (top) and the two exclusive columns (left). We do find it in all these places (and never outside), with the exception of the 2>1 corner, where a generic *a-* marker is used instead:[5] van Driem (1987: 77ff) interprets it as an indeterminate first-person marker. Nevertheless, uninflectedness for clusivity in this area is puzzling; precise discussion will be delayed until the analysis in Section 4.3.

Additionally, some markers are generalized compared to where they were in the intransitive paradigm: the *-ŋ* suffix (as a part of *-aŋ*) was present only in 1SGS.PRET cells. Now, it is found in 1SGA (topmost row) regardless of tense, and 1SGP.PRET (leftmost column) scenarios, too; but also in non-preterite scenarios 1SG>2 and 1SG>3.AFF. It exemplifies a robust form-to-function association with 1SG expression. However, note that the function-to-form mapping does not always make use of *-ŋ*: it is not found in 1SG>2SG scenarios, even though it could be conceived there as evidenced by the 1SG>2NSG cells.

Finally, some other markers are introduced that have no counterpart in the intransitive paradigm: notice the conspicuous presence of an *-u* marker in the last three columns. This marker is a clear exponent of third-person patient, as it is found almost everywhere there, and nowhere else. It combines readily with dual number marker *-si* to form *-u-si* (3DUP), as well as with other person markers (*kɛ- -u* 2>3). Note that unlike *kɛ-* and *-ge* above, which indifferently express an A, P or S, *-u* is role-sensitive: it is used exclusively for *patient* third persons.

In sum, though it presents several dozen cells, Limbu conjugation stands out as a fairly tame system, with recurrent, clearly analyzable parts. The markers that do find

[5]The *gɛ-* marker in these forms is a prefix, allomorph of *kɛ-*; not to be confused with suffixal *-ge*.

exceptions, like *-ŋ*, still have an overall coherence: they carry a definite content independently of the surrounding material, in contrast to more elusive morphomic markers of other languages. Aiming to encode this coherence in a formal model should make us lean towards morphemic and concatenative approaches to morphology whenever we can. Nevertheless, accounting for the details outside of this coherence does represent a challenge for a morphological theory of inflection.

3.1.3 Peculiarities

Let us now turn to distortions that can be found in the otherwise robust agglutinativity of the system. Four types can be identified. First, disruptions in the agglutinative template, and the "one property, one slot" principle. Second, the fusion of certain properties into cumulative exponents, of several different kinds; and more generally non-canonical exponence. Third, syncretism of markers above and beyond their meaning in other parts of the paradigm (canonical syncretism). Fourth, total collapse of distinctions that could otherwise be made with little apparent functional cost (uninflectedness). It is worth stressing again that these phenomena constitute distortions in limited zones of the paradigm, or of the template. Although most cells need to take one of these phenomena into account for some subpart of their form or their meaning, the other subparts will largely be unaffected in any given cell.

Limitations of the template

As could be noticed already in Tables 3.1 and 3.2, markers that express different values of a property often do not align. This is apparent in person marking: the "tame" person exponents *a-* and *kɛ-* are prefixes, but one also finds *suffixes* that clearly aim to express person, like *-u* and *-ge*. The same is true of number of the agent, which in >3 scenarios may be marked before or after *-u* (***-s**-u* DU>3 vs. *-u-**m*** PL>3). Negation is almost always expressed with a prefix and a suffix, but in cells with *-paŋ* or *-mʔna* no negative suffix is found (note [a] of Table 3.4). Thus, assigning a property to invariable slots can be difficult, and the converse is also true. There are no formal criteria to distinguish the position class of reflexive *-siŋ* (and its allomorph *-nɛ*) on the one hand, and that of the 2>1 marker *-nɛ* on the other: they are always adjacent to the stem, so that no marker can break the parallelism between them.[6] van Driem (1987) himself concedes to use a single slot for them.

His methodology, which tries to make slots and properties correspond one-to-one, to the point of distinguishing "non-singularity of agent" as opposed to "duality of agent", shows that some markers with different values of the same property cannot be made to share position classes. In this way, Limbu departs significantly from agglutinative languages with more rigid templates such as Swahili. On the whole, position classes

[6] For instance if there was a marker *-X* such that it followed *-siŋ* (*-siŋ-X*) but preceded *-nɛ* (*-X-nɛ*).

are better considered attributes of individual markers rather than of whole properties. This is the approach taken in this work (already preceded in this by Stump 2017), which I believe can help in formulating more generalizations.

Fusional marking

A case where assignment of properties to distinct slots fails especially is with fusional markers. When compared to Indo-European or Bantu, one might expect fusion to follow certain patterns. Indeed, some such patterns are found: despite the general decorrelation of person and number noticed in §3.1.2, some combinations of features receive special, non-compositional exponence. Moreover, this may include fusional marking of person–number with tense, or polarity; the former reminiscent from Romance languages (French *-ont* 3PL.FUT), the latter from Swahili (*si-* 1SG.NEG). Such an atom of form is the 3PL marker *mɛ-* (Table 3.1), which cannot be construed as a mere plural marker (despite zero third-person exponence) because *-i* already fills that function in most intransitive scenarios. It is able to express both plural number and third person. This is also the case of "submorphemic" *-ŋ* (as occurs within *-u-ŋ*, *-aŋ*, *-paŋ*), which is always associated with first person and singular number conjointly. As for other markers conditioned by, or even expressing, tense and polarity, the intransitive paradigm (seen in Tables 3.2 and 3.3) also foreshadowed markers such as *-ʔɛ* 1SG.NPRET and *-mʔna* 1PE.PRET. Cumulation with negation can be seen in the 1SG(S/>3).PRET.NEG cells of Table 3.4, with *-paŋ* being an exponent of 1SG.PRET.NEG.

A more surprising fusional phenomenon can occur, which is unexpected when coming from systems where marking of either participant in transitive scenarios is decorrelated: simultaneous expression of both participants by an atomic marker. Indeed, Swahili always separates A and P marking; as does French, on different sides of the verb. Suffix *-nɛ* 1SG>2SG is a clear example of this: it does not make use of *kɛ-*, marking second person, nor of any first-person-singular material seen elsewhere. Seeing its extent across the 1>2 corner of the paradigm, we can see that it does not quite encapsulate all four possible participant features in the 1SG>2SG scenario. Actually, it acts as the provider of exactly two pieces of information: the agent is first-person, and the patient is second-person; information otherwise separated. This is the definition of a portmanteau marker, and this portmanteau marker expresses 1>2. As Stump (2022: 172f) points out, the *a-gɛ-* marker for 2>1 can also be seen as expressing specific roles, in addition to the role-insensitive person information that either its submarkers contributes. However, I will not consider this exponent atomic or even a combination with emergent content: it will be introduced by two entirely independent rules in my IbM analysis.

These constitute distortions in a neatly arranged template where different properties make use of different markers. However, these cumulative markers do tend to follow Pāṇinian principles: they replace the more general markers that could occur along-

	P<3SG	S	A>3SG
1SG	-ʔɛ	-ʔɛ	-u-ŋ
2SG	kɛ-	kɛ-	kɛ- -u
3SG	-u	∅	∅ -u ∅

(a) In the singular

	P<3SG	S	A>3SG
1DE	-si-ge	-si-ge	-s-u-ge
1DI	a- -si	a- -si	a- -s-u
2DU	kɛ- -si	kɛ- -si	kɛ- -s-u
3DU	-u-si	∅ -si ∅	∅ -s-u ∅

(b) In the dual

Table 3.5: Zero exponence of third person

side them. Suffix *-mʔna* 1PES.PRET prevents all three markers from appearing: *-ɛ* PRET, *-i* PL and *-ge* 1EXCL. These isolated but robust markers constitute one argument for an inferential–realizational theory of inflectional morphology: any incremental theory, conforming to a "one property, one morph" principle, will struggle formulating complex combinations such as the ones seen in these markers.

Other non-canonical exponence

As we have already noticed, some properties are not formally marked in any slot at all. van Driem (1987), and my own Table 3.1, decide to assign "significant zeros" to particular slots, but of course this is only really meaningful in the case of full featural coherence of a given slot. Indeed, zero exponence is first and foremost determined by paradigmatic contrast, and the concept makes sense regardless of whether its contrasting exponents all lie in the same syntagmatic position. Let us recall the instances of zero exponence in Limbu in greater detail. Affirmative polarity and non-preterite tense are easy to notice as against negative and preterite, and have been mentioned already; their zero exponence stands out as quite uneventful in cross-linguistic typology. Other dimensions have more than simply two values, and zero exponence there requires more conspiracy in the system (several values must be positively marked instead of just one). For instance, the third person of A or S is not signalled by presence of any marker (except cumulative *mɛ-*). Rather, its forms contrast with other forms with actual person exponents. The most common such exponent is *kɛ-* 2, because it is transparently a person marker, not cumulative with any number feature. However, the picture is still quite explicit in Table 3.5: no special form is used for third person in S/A roles.

Note that, in the transitive case, one must choose an interacting participant; because of the reflexive-less nature of the system, a third person participant is the case where most contrasting cells exist. When instead, we choose a speech act participant (first or second person), the specific properties of person marking in the "local" scenarios as opposed to the third-person-involving scenarios mean the contrast is not as crisp as in Table 3.5. Even still, we can almost illustrate it with *a-gɛ-* 2>1[NSG] vs. *a-∅* 3SG>1PI. Thus, no marker is specific to third person singular/dual S/A participants, but cells involving them are still successfully distinguished thanks to paradigmatic contrast: this is zero exponence of third person.

The same can be noticed for singular number, whatever the function (except for first person which is always cumulatively marked). The formal marking of dual with *-si* and plural with *-i/-m/mɛ-/-si* (depending on the specific person and role) means that singular forms can still be identified. Notice that this holds true whatever the role of the singular participant: to see this, simply consider the parallelism between *kɛ-* 2SGS; *kɛ-* 2SG<3SG; *kɛ- -u* 2SG>3SG (and the corresponding 2DU and 2PL forms). Therefore, there is also pervasive, role-independent zero exponence of singular number in Limbu. Interestingly, the non-singular numbers do not have such role-independent expression.

On the opposite end to zero exponence is multiple exponence, when several cues, or exponents, together serve to express a given meaning. This has been noticed for negative marking, with clear redundance between the two markers in a given form: *kɛ-**n**- -ɛ-tchi-**n*** in (1). Another, more subtle case is that of copying in 3NSGP cells: *-u-m-be* 1PE>3SG vs. *-u-m-si-m-be* 1PE>3NSG. It is an instance of type 1 multiple exponence, using the typology of Harris 2017.[7] Each marker is transparently an exponent of plural, but both are required together: this is multiple exponence of (plural) number.

When extendedness combines with fusion, we find so-called overlapping exponence. This requires a cumulative marker, and one feature of this cumulative marker is also expressed elsewhere (hence, multiply). The *-paŋ* suffix is unique to negative forms (Table 3.4), but does not preclude the usual negative prefix *mɛ-*: *mɛ-n- -paŋ*.[8] Moreover, it serves to express 1SG>3.PRET in addition to negation: this qualifies as overlapping exponence. Another fusional exponent we saw previously is *-nɛ*, expressing 1>2. Exponence of first person is not forbidden in presence of *-nɛ*, though. Rather, markers specific to first person but with additional information may well appear alongside it: *-n-i-**ŋ*** 1>2-PLP-1SG '1SG>2PL'; *-nɛ-tchi-**ge*** 1>2-NSGA-1EXCL '1NSG>2'. Both *-ŋ* and *-ge* carry first-person content. Note that this is more intricate but also more motivated than the previous case, given that the two markers involved have some featural overlap instead of featural inclusion of one inside the other. This also means that one is not entirely predicted by the other, and considerations of morphological completeness motivate the appearance of both. This is not the whole story, however: for greater detail on copying of *-ŋ*, see §3.4.

These instances where the form–function mapping is not one-to-one are indications that, as noticed by Stump (2001), a theory fit to describe Limbu conjugation should incorporate several components. First, it should be able to account for extended exponence, and this fact is fatal to incremental theories on parsimony grounds. Sec-

[7] The difference with type 2 is not quite apparent; indeed the carrier morpheme *-si* does not require the extra nasal exponent (of number/negation) in any way, outside of these forms. However, other typical characteristics of type 2 might rule this instance out.

[8] It even requires another negative prefix instead of the usual suffix, but this is orthogonal to our discussion in that *-mʔna*, a polarity-independent marker, also triggers this phenomenon.

ond, a notion of paradigmatic contrast should be implemented to deal with zero exponence, instead of postulating zero morphs in the many places where they are needed. Although Limbu does not present any significant non-concatenative morphology, these facts are most naturally present in a Word-and-Paradigm theory of inflection; what is called, since Stump 2001, an "inferential–realizational" theory.

Syncretism

In contrast to the neatly distinguished cells in the intransitive paradigm, syncretism is a pervasive part of the Limbu transitive system, as can be seen from the many arrows in all corners of Table 3.4. This means that despite the availability of different markers, distinctions are not made everywhere they could be. Two families of syncretic cells involve only partial collapse of distinctions: these are the bottom and right ones in Table 3.4. They identify forms of third-person scenarios that involve non-singular number. However, they keep the singular/non-singular divide by using zero exponence for the former and some marker for the latter; this is canonical syncretism as defined by Baerman, Brown & Corbett (2005).

The non-singular marker in either case is particularly interesting. In the right-hand columns, the marker *-si*, otherwise pervasive in a person-independent dual meaning, receives a non-singular interpretation. This can be seen by systematically comparing the 3NSGP cells with the 3SGP ones: the contrast that emerges is *-si* NSG vs. ∅ SG. This can also be noticed in the 1NSG.EXCL>2 cells (*-nɛ-tchi-ge*): while 1NSG.EXCL>3 cells happily distinguish dual and plural numbers of the agent, the contrast collapses with a second-person patient. This is quite a special case as *-si* may contrast with a number of different markers in other contexts: *-m* for plural agent number in NSG>3 scenarios; *-i* for plural patient/subject number in NSG<3 and NSGS; *mɛ-* in third-person plural subjects. In the 3NSGP cells, however, no *mɛ-* is available; and in 1NSG.EXCL>2, no *-m* is available. By paradigmatic contrast, the meaning of *-si* becomes NSG. van Driem (1987: p. 31) hypothesizes that, for *-si*, this is the result of an "over-generalization of the original dual sense to include a notion of plurality". This can be witnessed outside the verbal paradigm, even outside third-person: numerals 2 to 9 are suffixed in that way (p. 32); copular suffixes display the same behavior in the first person (p. 56).

Consider now the bottom rows of Table 3.4, i.e. the cells with a third-person A. While *mɛ-* is unavailable for 3NSGP, we find that conversely *-si* is unavailable for expressing dual in some 3NSGA cells. There, we find *mɛ-* 3PL (and its allomorph *m-*), which we have already seen in the intransitive paradigm as the exponent of 3PLS, in dual cells. This substitution of *-si* with *mɛ/m-* does not happen in all 3NSGA cells, however. Notice that, in 3NSG>3 cells, *-si* is used within the recurrent *-s-u* combination used for all DU>3 scenarios.[9] Therefore, it is only in special inflectional contexts that *mɛ-* is licensed

[9] Some special conditions may motivate use of *mɛ-* even in such dual scenarios: see van Driem (1987)'s p. 84 examples. This is a general property of "plural" *mɛ-*, however, as this happens in the intransitive

to have a general meaning.

These two instances of syncretism disrupt to a degree the "clearly identifiable" properties that *mɛ-* and *-si* express otherwise. Some contexts license a restrictive 3PL/3DU interpretation, while others are happy with a laxer use for just any non-singular participant (provided its features comply with the syncretism conditions). Then, while these markers invariably contribute some flavor of non-singular, their precise contribution is highly dependent on the general inflectional context, and in specific corners they can even contrast by expressing incompatible values of number. This can be seen as a distortion of the agglutinative patterns of the system, whereby each word can be neatly cut into morphemes with constant contribution. In fact, the numerical pervasivity of third person certainly highlight this variability as a force to be reckoned with in the Limbu system.[10] This undetermination of the global meaning by the individual components of the word is precisely a phenomenon that Stump (2001: 7f) identifies as difficult for incremental theories of inflection.

Uninflectedness

In addition to partial collapse of some distinctions, total uninflectedness for certain features is also present in the Limbu system. A clear case can be noticed in the 2>1 corner of the paradigm of Table 3.4. While distinctions are represented with different text, identity of form uses arrows, and this nine-cell square only features two different forms (if one ignores tense allomorphy; or if you restrict to just one tense and polarity value combination), much fewer than the usual six in other 3-by-3 squares of the paradigm. More specifically, this form is entirely insensitive to number of either participants: it receives no additional number marking whatsoever, even though such number marking is certainly available in principle (if only *-si* DU and *-i* PL for the first-person P). In fact, number is not entirely collapsed as it is sensitive to singular number of both participants. Notice, however, that as soon as either one of the participants is non-singular, the number distinctions for the other become irrelevant: it is only when one of the participants are singular that sensitivity is possible at all. We are not in presence of total uninflectedness of both number features together, but certainly conditioned uninflectedness in either. Moreover, the uninflectedness is implemented via the absence of marking, in ways reminiscent of German non-genitive marking (see the WAGEN paradigm in Table 2.1b). This is in contrast to the previous case of syncretism, where some exponent of number remained. Further, notice how this is better not analyzed as a case of zero

paradigm, too.

[10] This means that third person is somewhat less differentiated in non-singular numbers than first and second persons. This property of neutralization is interpreted by van Driem (1987: 270f) in terms of well-known animacy/referential hierarchies, citing Comrie (1981). That the collapse remains uneffective in 3NSG>3 configurations seems to be in line with either lesser markedness or more common use of the third person.

exponence of number, seeing as no paradigmatic contrast is available that would give a definite meaning to the absence of a marker.

A more subtle case can be found in the 1>2 area of the paradigm; the form *-nɛtchige* is found for all cells with a non-singular agent and any second-person P. In the "agent number" dimension, it only partially collapses number, as it contrasts with singular agent number. However, in the "patient number" dimension, no distinction is made whatsoever. One could conceive of phonological phenomena that would analyze these cells as underlyingly inflected for number (for instance, haplology to **/-nɛ-tchi-si-ge/* 1NSG>2DU, or vowel coalescence to **/-nɛ-tchi-i-ge/* 1NSG>2PL), but no independent evidence can be given for such forms, and these hypotheses do not bear upon the necessary disambiguation of the surface form by the listener. Therefore, it is more parsimonious to assume that these forms are uninflected for patient number. This prompts the distinction, despite formal similarity, between *-tchi* in the 1SG>2DU cell, which expresses dual number of the patient; and *-tchi* in the 1NSG>2 cells standing for non-singular number of the patient. (This recapitulates van Driem (1987: 8X?)'s analysis.)

It is worthwhile noticing the unexpectedness of such forms, from an agglutinative standpoint. In an otherwise agglutinative language, the fact that some combinations of TAM markers do not occur seems understandable in view of the great semantic intricacies of this domain: the combination of semantic content simply does not make sense, or less sense. By contrast, here we find incompatibilities which seem to lie in the domain of morphology itself: it is not that the particular content is not conceivable, nor that it does not have a corresponding form; rather, the morphology chooses not to use certain markers in the form. This might be the result of fossilized socio-pragmatic pressures, as explained by DeLancey (2018), but the sharp judgment facts (see van Driem 1987: pp. 78–79) indicate that this is fully grammaticalized, and ought to be accounted for by our morphological theory.[11]

3.1.4 Type of exponence and formal morphological theory

As a summary of the data seen so far, let us recall a few key points about the Limbu system:

- **Discrete markers for person and number**

  Limbu separates person and number inflection for a given participant into different markers.

- **No featural coherence**

  Markers realizing different values of the same property are often realized in different surface positions.

[11] See Evans (2000: p. 107) on local participant prefixes in Iwaidjan.

- **Limited exponence of syntactic role**

  Syntactic role is not always expressed on person and number markers.

- **Fusional characteristics** (Matthews 1972)

  The system still exhibits many disruptions in a generally agglutinative pattern, which are characteristic of fusional systems, notably in the form of non-canonical exponence.

- **Conditional syncretism**

  A few markers have a core meaning, but determining their precise function requires information that can only be found in the rest of the form; that is, individual elements of form sometimes underdetermine function.

- **Uninflectedness**

  Some relevant inflectional information may not be encoded overtly: either it is resolved by means of paradigmatic opposition, or not at all, leaving us with residual ambiguity.

These last two points are worth some further discussion. On the one hand, the form can have greater function than the sum of its parts: markers can acquire some additional, local function by paradigmatic contrast. This holistic behavior of markers complicates the form-to-function mapping by "enrichment". However, the independent existence of each marker suggests a partly atomistic view, where meaning is not entirely emergent. Conversely, the function-to-form mapping can suffer impoverishment (different functions give rise to the same form), because some markers that could appear are not able to, for reasons that are unclear as they would not be redundant.[12] A form such as **-nɛ-tchi-si-ge* expressing 1DE>2DU seems conceivable, but happens not to exist. The absent markers may not be replaced at all, or replaced by a single other marker which thus exhibits syncretism. In all cases, these are textbook examples of syncretism, caused by a combination of features instead of specific inflectional properties of lexemes (Baerman, Brown & Corbett 2005). This syncretism is sometimes counterbalanced by the holistic meaning that syncretic forms may acquire, just like above, but even in forms with "lacking" elements.

Two components of our morphological analysis are suggested by this tension between atomistic and holistic exponence, giving rise to unexpected local behavior of forms. These are: an ability to use paradigmatic contrast to enrich the meaning of single markers, or even full forms, on the one hand; and a powerful ability to generalize

[12] However, different forms giving rise to the same function would be "overabundance", which is found at several places in the system (see the 1SGS cell of Table 3.4; and *-u(n)sin* NPL>3NSG.NEG forms in Table C.1).

over instances, on the other. In case paradigmatic contrast cannot capture finer details, our knowledge representation system should allow each instance to retain its potential for idiosyncrasy.

Having surveyed several types of phenomena that occur in the Limbu system, we will now turn to more specific problems that arise there. This is especially the result of intricate distributions. In the next section, I explore the distributions of interest in more detail, to carve out similarities and differences between them, and the related challenges for an analysis of Limbu conjugation.

3.2 Discussion of the data: specific problems

Now that the general tendencies of Limbu inflectional morphology and a basic typology of morphological splits have been described, let us turn to the challenges that specific markers pose for the analysis of Limbu. The reason why they cannot be straightforwardly analyzed is often because their clean distributions (like those of Table 2.1) find exceptions that do not quite reduce to Pāṇinian competition. These exceptions come in different varieties: they can (i) take the form of a specific competitor, (ii) take the form of a competitor which has other purposes elsewhere, or (iii) they can be characterized by the absence of marking. However, the naturalness of the overrides does allow one to draw quite detailed inferences on the function of the defaults thanks to paradigmatic opposition. Therefore, while strictly speaking, they do follow a morphomic pattern, a bird's-eye view encourages us to try and encompass their natural- and Pāṇini-like behaviors.

Three cases will be examined. The first features quite a straightforward, specific override (type i) in the first plural exclusive preterite, which surfaces indifferently in two areas, usually divided by a split: this will be the canonical case of what we will call a "pseudo-Pāṇinian" split. Next, partial neutralization of number distinctions in the third person, which differs depending on role, provides a split with two interlocking L shapes, a textbook example of bidirectional syncretism (Stump 2001: p. 212). Because the syncretic markers have other purposes than as simple overrides (type ii), the precise analysis will require a closer look at the function of number markers outside the third person. Finally, the 2>1 part of the paradigm will be looked at in more detail, to delineate the specifics of the competition of first-person markers and of their conspicuous absence in certain areas of the paradigm (type iii).

3.2.1 First plural exclusive

One of the few tense-sensitive allomorphs found in the system is the preterite first plural exclusive marker *-mʔna*. It occurs in the intransitive paradigm of Table 3.3, and pre-

Role →	A>3SG	S	P<3SG
1PE.PRET	**-mʔna**	**-mʔna**	-i-ge
1PE.NPRET	-u-m-be	-i-ge	-i-ge
2PL.PRET	kɛ- -u-m	kɛ- -i	kɛ- -i
2PL.NPRET	kɛ- -u-m	kɛ- -i	kɛ- -i
1PI.NPRET	a- -u-m	a- -ɛ	a- -ɛ
1PI.NPRET	a- -u-m	a-	a-

(a) First plural exclusive split

	−	+
α	**A**	**A**
β	B	C
γ	B	C

(b) A general pseudo-Pāṇinian split

Table 3.6: Pseudo-Pāṇinian splits

empts the surfacing of preterite *-ɛ*,[13] plural *-i* and exclusive *-ge*. This is quite expected given a mechanism of Pāṇinian competition, because *-mʔna* serves to express more specific content than any of these three markers; it appears as a portmanteau marker. Note that *-mʔna* is specifically excluded when the first person is a P and not an S: *-i-ge* is the only form in 3SG>1PE regardless of tense. Therefore, while *-i* is indiscriminate between P and S roles, *-mʔna* is specific to S. This is still in line with an ordinary Pāṇinian pattern.

However, this portmanteau is not quite as specific as the example suggests. Indeed, *-mʔna* is found also when the first person is an agent, and the patient is third-person; that is, the 1PE>3.PRET cells. This time, the relevant markers that one should expect in these cells are *-u*, for third-person patient; and *-m*, for plural of the agent. The exclusive and preterite markers are unchanged except for place assimilation of *-ge* into *-be*; the combination of the first three is in fact found in the non-preterite counterpart: *-u-m-be* 1PE>3SG.PRET. The behavior of *-mʔna* as a portmanteau is not surprising because the other markers are so general, if not for the specific facts of competition. Then, to override *-u*, expressing 3P, and *-m*, expressing PLA, using Pāṇini's principle, we must say that *-mʔna* serves to express at least a 1PEA>3P function in these cells.

Paradoxically, *-mʔna* is able to surface both in S and A roles: this suggests abstracting the distinction away by underspecification, and making it a common marker to all scenarios in which it occurs. On the other hand, it surfaces at the expense of either *-i* or *-u-m*, both of which are incommensurate with it in terms of function: 1PE(A∨S) merely overlaps with PL(P∨S) and PLA.3P, without being more specific than either. The situation is summarized in Table 3.6a: the fact that the distribution of *-mʔna* is not more specific than *-u* or *-i* is translated by the fact that the grey area straddles across the almost completely green column and across the almost completely red rectangle.

An additional representation, more faithful to the structure of the paradigm, is pre-

[13] Regularly elided before vowels in the rest of the paradigm; in particular, before *-i* and *-u* in other 1PE<>3 scenarios.

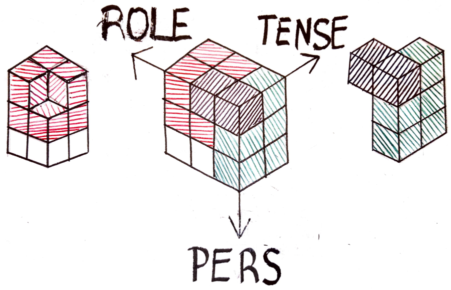

Figure 3.1: A 3D representation of the data in Table 3.6a

sented in Figure 3.1. The vertical axis of Table 3.6a combines information of tense and person (including clusivity), by interleaving the two. The axis is reorganized by spreading the list of combinations into two cross-classifying axes in Figure 3.1: the vertical one and the northeast-bound one. This represents the data in a 3 × 3 × 2 parallelepiped instead of a 3 × 6 rectangle, making explicit the fact that *-u* (green) and *-i* (red) are tense-independent markers, while *-mʔna* is specific to one value of tense, one value of person and two values of role. For clarity, I only represent the inflectional split at hand with colors, and ignore the full forms. Some cells are made invisible by the perspectivperspective but in this case the angle is chosen so that no exceptions are found in the hidden cells. In general, my feeling is that such a representation gives a one-glance, compact and intuitive sense for the distributions of each marker. Naturally, this mainly works for multi-dimensional systems with quite general markers.

A more schematic representation, which aims to serve as a model for later examples as well, is presented in Table 3.6b. Again, marker *A* overrides the split between *B* and *C*, without being more specific than either. The fact that the split between *B* and *C* occurs in more than just one pair of cells is paramount to the definition: the respective natural classes that *B* and *C* could claim are more than just one cell; and this is why *A* is a bona-fide override which prevents the distribution from being natural. It does not have quite the specificity that a rigid Pāṇini's principle requires for usual overrides, however. Because both *B* and *C* emerge as elsewhere cases, I will call such situations pseudo-Pāṇinian.

In this specific case, there are, however, good reasons to consider *A*, i.e. *-mʔna*, an override. First, notice that its status as a portmanteau can immediately account for the preemption of all other markers,[14] as a simple consequence of general Pāṇinian competition. Therefore, the high specificity that results from its fusion encourages a view where it can serve to express many functions instead of other markers. This maintains a

[14]Moreover, this elimination is *discontinuous* (in the sense of Noyer 1992) in the case of *-ge*, as evidenced by the contrast between non-preterite *-mʔna-si* and preterite *-u-m-si-m-be*: the morphotactics *-si* witnesses the separation of *-mʔna* and *-ge* into two different slots.

natural description of all markers with which *-mʔna* competes. Further, the ambiguity between an A marker and an S marker, which speaks against two fully cell-specific descriptions, is in fact quite spurious. Most lexemes will only be transitive or intransitive, and *-mʔna* will only occupy a single cell in their paradigm, so that virtually no use case will feature genuine ambiguity. Within each lexeme's paradigm, *-mʔna* will indeed be more specific than *-u-m*, or than *-i*.

Therefore, keeping in mind these two facts, I will argue for an analysis which recognizes an abstract *-mʔna* marker, with properties that are common to both cases; the actual use cases will in turn employ role-specific subtypes of this abstract *-mʔna*, a distinction which is justified by the valence of individual lexemes. This allows us to reap the benefits of an aptly designed competition device, while accounting for an abstract object above and beyond the use cases.

This configuration of knowledge representation can be regarded as an instance of the contrast between productive rules and subregularities drawn by Koenig (1999: p. 130): the presence of *-u*, *-m* and *-i* emerge as *productive rules* or patterns that are each common to a certain class of words, defined intensionally. The general rule for *-mʔna* uses the exact same formal mechanism, namely class abstraction. However, this marker *-mʔna* is characterized by a *subregular* pattern, because the intensional description that its general rule enjoys is closed down by an extensional enumeration of the members, i.e. the specific rules. This class enumeration is the defining characteristic of subregular patterns in Koenig (1999: p. 130)'s interpretation. In this way, abstract templates which are used for regular rules can also be put to use in subregular patterns, alongside some form of lexical storage.

3.2.2 Dual and plural third persons: *-si* and *mɛ-*

At first glance, the three-way number contrast seems to be active in the third person: *-si* and *mɛ-* contrast as agreement markers with the intransitive subject; the former in a dual function, and the latter in a plural function. This natural split also applies when the third person is acting on another third, as evidenced by the vertical divide in the middle section of Table 3.7.[15]

In this context, the markers each express a specific number value: one would expect a natural distribution, with a vertical split across the paradigm. While this would be consistent with the recurrent dual function of *-si* in other persons, the presence of overrides in all P and some A roles mean that neither of the other vertical generalizations works: *-si* is used for all dual third persons except when they are agents acting on a non-third person; *mɛ-* is used for all plural third persons except when they are patients. Therefore, one finds two interlocking L shapes instead of two stripes. This is an

[15] Presumably this reflects the original contrast that existed between these forms, given the formally and functionally similar markers in other languages (e.g. Camling *mi-* 3PLS; Thulung *-mi* 3PL), and that non-singular *-si* is described as a "generalized dual" by van Driem (1987).

↓ Role	3SG	3DU	3PL
A>1/2	∅	**mɛ-**	mɛ-
A>3	-u	-s-u	mɛ- -u
S	∅	-si	mɛ-
P	-u	-u-si	-u-**si**

Table 3.7: Markers of third person

example of divergent bidirectional syncretism, in the terminology of Baerman, Brown & Corbett (2005: p. 134), based on that of Stump (2001). This situation is tough to analyze in terms of the splits seen in Chapter 2, because each marker is either a default or an override, depending on the section in the paradigm. If anything, one could consider it a "balanced morphomic split".

Baerman, Brown & Corbett (2005: pp. 139–145) study a handful of similar patterns of syncretism, and argue that symmetrical rules fail to account for the data in general, "requiring that some element be treated as accidental". Unlike the instances accrued there, I shall contend that a powerful generalization tool can treat this example of divergent bidirectional syncretism with great detail, and that good arguments can be found for treating *mɛ-* as an exception.

The presentation in the previous paragraphs would tend to outline *-si* as the elsewhere case when it comes to role: there is no straightforward way of including all roles except "A>1/2" cells by means of a natural description. On the other hand, as a first approximation, *mɛ-* could be analyzed as a "nominative" non-singular marker (i.e., in S or A roles). However, treating *-si* as the elsewhere case forces *mɛ-* to be unavailable where the dual–plural contrast is maintained; this case can only be excluded by having one rule with a disjunctive distribution, or two formally identical rules with different, overlapping meanings. An inheritance-based knowledge representation mechanism can allow generalization across rules in the latter case.

The analysis of *-si* as a general, "elsewhere" non-singular marker finds yet more support in the rest of the paradigm. Parsimoniously, the L-shaped distribution of *mɛ-* can be taken at face value as a disjunctive pattern, because it is restricted to the third person – all use cases are found in Table 3.7. The dual-like *-si* marker, however, has both a much wider spread across the paradigm, always serving to express similar functions, and a more intricate pattern (see Table 2.3). The crucial observation is that every use case in a dual function finds a contrasting, specifically plural marker. For instance, in first and second persons, while markers *-m*, *-i* and *-mʔna* can all be used to express plural (cf. Table 3.6a), *-si* contrasts in a dual function with all of these specific plural markers. If we accept *mɛ-* as an essentially disjunctive marker, then treating *-si* as a more general allows this occasional dual function to be an emergent property of the other markers

↓ Role	1SG	1DE	1PE
A>2SG	-nɛ	-nɛ-**tchi**-ge	
A>2DU	-nɛ-tchi-ŋ	-nɛ-**tchi**-ge	
A>2PL	-n-i-ŋ	-nɛ-**tchi**-ge	
A>3SG	-u-ŋ	-s-u-ge	-u-m-be
A>3NSG	-u-ŋ-si-ŋ	-s-u-si-ge	-u-m-si-m-be
S	-ʔɛ/aŋ	-si-ge	-i-ge
P<3SG	-ʔɛ/aŋ	-si-ge	-i-ge
P<3NSG	mɛ- -ʔɛ/aŋ	mɛ- -si-ge	mɛ- -i-ge

Table 3.8: Markers of first-person number

and paradigmatic opposition. This is in line with the collapse of distinctions which is found not only in 3P contexts, but also in 1>2 scenarios: it is visible in Table 3.8. Again, whenever a dedicated plural marker is unavailable, *-si/-tchi*[16] surfaces as a general non-singular marker.

Therefore, in certain unclear patterns of split, generalizations that can be formulated for one competitor but not the other can arbitrate in favor of a particular line of analysis. In this way, we are able to resolve in similar ways the competition that *-mʔna* is subject to, and the competition between *mɛ-* and *-si*, by identifying the less general marker and tailoring the analysis to its distribution. This can be justified by a more exemplar-based memory of exceptions; but does not prevent the emergence of properties for the elsewhere case. When the distinction can be drawn, this puts us into a position where we can formulate a more parsimonious analysis of both members of the competition.

3.2.3 Uninflectedness

As was noticed in Section 3.1.3, there are no contrasts in number in the 2>1 area of the paradigm.[17] This differs from the intransitive paradigm and those parts of the transitive paradigm where first or second person interact with a third person (this draws out a mirrored L-shape with its corner in the bottom right of Table 3.4). In all of these areas, not only are there number contrasts, but two different markers are available to ensure them. No number markers whatsoever appear in 2>1 scenarios however: notice the disruption of the vertical split in the bottom rows of Table 3.9. Rather, two person prefixes are found: *a-* for first person (also used in first-person-inclusive cells) and *kɛ-*

[16] Recall from page 9 that the alternation is phonologically conditioned.

[17] Or rather, under the condition that one of the participants is non-singular, there are no number contrasts for the other.

↓ Role	1SG	1DE	1PE
A>3SG	-u-ŋ	-s-u-ge	-u-m-be
A>3NSG	-u-ŋ-si-ŋ	-s-u-si-ge	-u-m-si-m-be
S	-ʔɛ/aŋ	-si-ge	-i-ge
P<3SG	-ʔɛ/aŋ	-si-ge	-i-ge
P<3NSG	mɛ- -ʔɛ/aŋ	mɛ- -si-ge	mɛ- -i-ge
P<2SG	kɛ- -ʔɛ/aŋ	**a-gɛ-**	**a-gɛ-**
P<2NSG	a-gɛ-	**a-gɛ-**	**a-gɛ-**

Table 3.9: Markers of first-person number: different area (the top part overlaps with Table 3.8)

(allomorph *gɛ-*) for second person. Besides, notice how the exclusive person marker *-ge* is not used either. Any analysis which recognizes number markers elsewhere should be able to account for their absence as well; especially if it postulates very general markers such as non-singular *-si*.

As we can see in Table 3.9, the marking of patient number is suppressed, but this is also the case for number marking of the agent. This symmetry is witnessed by the pattern at the top of Table 3.10, which mirrors that at the bottom of Table 3.9. To account for the absence of either of these two types of number marking, a line of analysis analogous to that suggested for *-mʔna* in Section 3.2.1 above does not seem appropriate. Indeed, it would need to postulate two zero overrides in this zone: one for agent number marking and the other for patient number marking. Moreover, to get the competition right so that *-si*, *-i* and *-m* are overridden, one would essentially need one override for each number value. This does not seem desirable. What is more, the properties of inheritance in typed feature structures entail that one cannot specify that each specific number value has an override without listing the values in question. One would arrive at a paradoxical situation which requires maximally specific zero overrides, as a solution to account for uninflectedness, which is the essence of unspecificity!

Similar facts characterize the 1>2 area. For person marking, instead of the conjunction of two person markers, a dedicated portmanteau marker *-nɛ* is used in all cells, thus expressing 1>2. The difference for number marking lies in that number is really distinguished only for the first-person agent: non-singular *-tchi* contrasts with singular *-ŋ*; and exclusive *-ge* is now available again. On the other hand, the second-person agent only contrasts for number when the agent is 1SG: this is represented by the two bottommost rows of Table 3.10. This time, an analysis should be able to reflect this asymmetry between first-person agents and second-person patients.

The exceptionality of these cells is quite clear in Limbu; but other related Kiranti

↓ Role	2SG	2DU	2PL
A>1NSG	a-gɛ-	**a-gɛ-**	**a-gɛ-**
A>1SG	kɛ- -ʔɛ/aŋ	**a-gɛ-**	**a-gɛ-**
A>3SG	kɛ- -u	kɛ- -s-u	kɛ- -u-m
A>3NSG	kɛ- -u-si	kɛ- -s-u-si	kɛ- -u-m-si-m
S	kɛ-	kɛ- -si	kɛ- -i
P<3SG	kɛ-	kɛ- -si	kɛ- -i
P<3NSG	kɛ-m-	kɛ-m- -si	kɛ-m- -i
P<1SG	-nɛ	-nɛ-tchi-ŋ	-n-i-ŋ
P<1NSG		**-nɛ-tchi-ge**	

Table 3.10: Markers of second-person number

languages also vary significantly in precisely these cells; namely, in what number they express at all, and if so, how much. Two examples will suffice: Athpare (Ebert 1997a) and Camling (Ebert 1997b) remove any distinctions in number of the first-person agent in 1>2 scenarios; precisely where Limbu keeps them. Second, while Limbu foregoes number marking entirely in 2>1 scenarios, Athpare can mark number of exactly one participant; either the object or the subject, giving rise to overabundance (Thornton 2011). Thulung (Lahaussois 2020), on the other hand, marks both together; with some exceptional collapses.

My analysis of the distributional facts of Limbu, in Section 4.3.2, will make crucial use of two facts: first, all number markers with similar functions, including *-ge*, pattern together in their appearance or disappearance. Second, these markers do not acquire any special function when they do surface, even though their presence is paradigmatically opposed to their absence; this suggests that the markers are conditioned on certain contexts without serving to express information about the context. These two facts find natural analytical devices in the IbM formalism. Systematic alternation can be successfully encoded in typed feature structures; a content/context distinction fruitfully accounts for conditions on the appearance of markers.

3.3 Implications of the data: the morphome debate

A general observation that emerges from the discussion in the previous section is that Limbu features an unusual type of split. Taking the definitions by the letter, these splits do not seem to reflect any phonological or syntactic motivations. Indeed, they cannot be considered natural because they find locally specific exceptions. Pāṇinian competition cannot be construed to apply then, either, because the overrides do not express strictly more than the defaults, if generalizations are formulated between them. Thus, the splits are morphomic (Aronoff 1994, Luís & Bermúdez-Otero 2016): while they cannot be said to serve no purpose outside of morphology itself, they do not correspond in the most straightforward way to requirements of syntax.

Therefore, it might be tempting to compare these splits to established morphomes discussed in the literature: for instance, the shape they take is reminiscent of L/N-pattern morphomes in Romance from Tables 2.2a and 2.2b. This is warranted inasmuch as it brings to light yet more evidence that morphology can have some autonomy. However, the behavior of these morphomes is quite different from the stability of Romance verbal ones: the impressive variation, both fine-grained and system-wide, documented in the neighboring Bantawa for participant marking (Cho 2020), suggests that idiosyncratic changes can easily take place in these splits.

Conversely, this variability seems to be related to the fact that these splits display some featural structure that speakers can pick up on. That is, they do not define such incoherent sets of cells as gender markers in Batsbi, seen in Table 2.2c: "unmotivated" would be much too strong a term.

Instead of having isolated stipulations in the lexicon, we can try to capture whatever morphological unity we can find using the tools that morphological theory provides for us. This approach asks the question that any formal theory should be concerned with: what devices are needed to account for the linguistic phenomena we find? While feature remapping has been proposed (Stump 2016, Bonami & Boyé 2008), it sidesteps this question entirely, without first considering other hypotheses. This does not seem justified in Limbu.

Therefore, I will try to account for the splits using just the tools that our morphological formalism has. They display conspicuous overrides, and I will accept this at face value in my analysis, and try to consider pseudo-Pāṇinian splits as a subclass of Pāṇinian splits. This will allow us to explore what it takes for a well-understood theory of blocking to work for these unusual cases. The hope is that my approach will help reduce pseudo-Pāṇinian splits to an established case of split; or at least, it will have raised awareness of intermediate possibilities that morphology can have between full Pāṇinian splits and clear-cut morphomes.

3.4 Dependency and copying phenomena

The extended exponence of number and negation in 3nsgP scenarios has been noticed already in Section 3.1.3. The markers involved in this phenomenon are *-ŋ*, marking 1sgA, and *-m*, marking 1∨2plA.[18] Let us first remark that they are not necessarily the

[18]The negative marker *-n* also occurs, but behaves quite differently: it only occurs secondarily, if neither *-ŋ* nor *-m* are present; and it is only optionally copied. I will not provide an analysis for this.

Carrier \ Dependent	*-m* 1∨2PLA	*-ŋ* 1SGA
-u 3P	✓	✓
-si DUP/NSGP	conditioned	✓
-i 1∨2PLP	✗	✓

Table 3.11: Occurrence of different carrier-dependent pairs in Limbu conjugation

result of copying, not in a phonological sense at least, since there may be just one instance of the morph in the form.[19]

Stump (2017: pp. 423–425) pointed out that they are special in two respects. First, they can occur in two different positions (i.e., “copied”), causing multiple exponence. This seems to be similar in some respects to Harris (2017) type 1 multiple exponence. Second, they are always preceded by specific material, and are entirely dependent on that material.[20] Only three morphs can serve as hosts, or carriers: *-u* 3P, *-si* DUP/NSGP, and *-i* 1∨2PLP; for instance: *-u-ŋ* 1SG>3SG.AFF; *kε- -u-m-si-m* 2PL>3NSG.AFF; *-n-i-ŋ* 1SG>2PL.AFF. Therefore, following Stump (2022: pp. 84–85), we will call this phenomenon morphotactic dependence. While Stump (2022) does identify that the dependents are both agent markers, an additional generalization can be made: the carrier always serves to express some property of the patient.

Stump (2022) noticed that, as a consequence of this morphotactic dependency, three cases occur: when no carrier is present, the dependent will not surface at all, as in *-nε* 1SG>2SG; when one carrier is present, the dependent occurs immediately after it, as in *-u-ŋ* 1SG>3SG; when two carriers are present, two pairs occur, as in *-u-ŋ-si-ŋ* 1SG>3PL. No cell features more than two carriers.

To be more specific, the different carrier-dependent pairs that are found are compiled in Table 3.11. A checkmark is found whenever the combination is obligatory, and a crossmark when it does not occur at all. In the case of *-ŋ*, any of the three carrier-dependent pairs can occur on its own: *-u-ŋ* as above; *-n-i-ŋ* 1SG>2PL; *-nε-tchi-ŋ* 1SG>2DU, *(mε-n-) -paŋ-si-ŋ* 1SG>3NSG.PRET.NEG.

For *-m*, however, the dependent–carrier pair *-si-m* cannot occur alone, and *-i-m* does not occur at all. There are two reasons for this. First, *-si-m* and *-i-m* are featurally compatible with 1PE>2NSG cells, just like *-si-ŋ* and *-i-ŋ* for 1SG>2NSG. However, the only number exponent that is found in such cells is the non-singular agent marker *-tchi*,

[19] van Driem (1987) considers *-ŋ* in *-nεtchiŋ* to be a copy of *-nε*, presumably on featural grounds.

[20] The hypothesis that *-ŋ* be also a dependent within *-aŋ* or *-paŋ* could be entertained. I choose not to pursue this possibility because the *-a* and *-pa* parts do not seem to have any independent existence: they will not be morphs in their own right in my analysis. We will still be able to capture the recurrent form–content association of *-ŋ* with 1SG, beyond its use as a dependent, in Section 4.5. This is thanks to HPSG and Online-Type Construction’s powerful abstractive powers.

	Suffix positions[21]						
A>P	1	4	5	7	8	9	10
1SG>2SG	nɛ						
1SG>2DU	nɛ	tchi_1	**ŋ**				
1SG>2PL	n(ɛ)	i	**ŋ**				
1SG>3SG		u	**ŋ**				
1SG>3NSG		u	**ŋ**		si	**ŋ**	
1NSG>2	nɛ			tchi_2			ge
1PE>3SG		u	**m**				be
1PE>3NSG		u	**m**		si	**m**	be

Table 3.12: From Stump 2022: p. 87

and no patient exponent is found. Seeing as the carrier morph in these pairs always expresses properties of the patient, no *-m* can surface in these scenarios. Second, PL>3NSG scenarios always feature an *-u-m* pair in addition to the *-si-m* pair. The only exception to this is with the fusional marker *-mʔna*, which serves to express 1PE>3.PRET, and overrides both *-u* and its dependent. In this case, the remaining non-singular patient marker *-si* does not host any dependent *-m* either: *-mʔna-si* 1PE>3NSG.PRET.

These facts could support a phonological copying analysis, where consonantal material is copied after non-singular *-si*, only when available. Instead, following the remarks of Stump (2022: pp. 94–5), I will adopt the view that the rules introducing the two carrier-dependent pairs apply independently of one another, and that the exception for *-mʔna-si* is stipulated. Apart from this unique exception, the original generalization by Stump that dependents surface whenever the carrier is present holds throughout the system.

One major difference with Stump's analysis, though, is in the conditions that govern the carrier-dependent rules. In his analysis, this carrier-dependent behavior is entirely determined by the slots, or groups, that the rules occur in Stump (2022: pp. 88–89). When potential carrier rules do not compose, another explanation must be found. For instance, *-tchi* NSGA in *-nɛ-tchi-ge* 1NSG>2 cannot host *-m*, and is instead relegated to another group. In the end, Stump is forced to introduce a three-way distinction, both in functions and in positional slots, between dual P/S *-tchi*$_1$*/si* (position 4), non-singular agent *-tchi*$_2$ (position 7) and non-singular patient *-si* (position 8). In that, he largely follows van Driem (1987). This leads to an analysis like that of Table 3.12.

My contention is that this introduces unnecessary distinctions in the analysis, so

[21] Stump (2022)'s suffix positions are identical to mine, found in Table 3.1, except that I do not recognize his slots 6 and 7 (partly inherited from van Driem 1987). Hence the two-slot difference in subsequent slot numbers.

that markers which could be analyzed together are artificially separated. van Driem (1997: pp. 171–172) argues that the non-singular functions of *-si* are found further from the stem because they are more recent innovations than dual *-si*. While this may be true, the only synchronic evidence for this is in 3NSGP *-si*. Thus, the choice to assign 1NSG>2 *-tchi₂* in any specific slot is arbitrary;[22] Stump's choice is only particularly convenient given the rest of his analysis of morphotactic dependence. Moreover, one could want to analyze *-tchi-ŋ* together with *-si-ŋ* despite their slightly different functions (1SG>DU vs. 1SG>NSG). My analysis will only keep distinctions between different *-si* markers when this seems motivated by the data, allowing for a simple two-way distinction based on role. I leave the details of this for Sections 4.3–4.4.

[22] van Driem (1997: p. 169) mentions this quite rightly, but goes on to use this as an argument for treating the non-singular markers *-tchi₂* NSGA and *-si* NSGP identically; my analysis will choose the other option, justified by a unified analysis of the agent markers.

Chapter 4

Analysis of the Limbu data

The facts that my analysis of Limbu will account for, and the broad approach that will be taken for each of those facts, have been presented. This section contains a brief presentation of the formalism that the analysis will be couched in, its specific powers, and the knowledge-representation tools it provides, in Section 4.1. In particular, inheritance and cross-classification in typed feature structure allow for a unified account of different patterns in Limbu conjugation.

Once the formalism has been presented, I will turn to building an analysis, recapitulating the facts seen in Sections 3.2–3.4. Different fragments of the system will be given, and enriched as the scope of the analysis widens. In the end, we will arrive at a full account of the affirmative paradigm; I review the main features of the analysis in Section 4.5.

4.1 The IbM formalism

In this section, I will present Information-based Morphology (Crysmann & Bonami 2016, 2017), an inferential–realizational theory of morphology, couched in a logic of typed feature structures, as used in HPSG (Pollard & Sag 1987). I will first introduce general knowledge-representation devices that are used throughout this chapter, because they are paramount to the architecture of IbM and analyses in IbM. Then, I will turn to specific assumptions and features of IbM, and how it implements generally recognized morphological principles.

4.1.1 Abstraction mechanisms

The main feature that IbM uses to represent the organization of linguistic objects is inheritance-based type hierarchies. They allow one to organize and classify objects in a straightforward and malleable fashion. Moreover, the generalizations that these classifications may express can be of many kinds. For instance, they can apply to virtually any zone of the grammar: lexemes, grammatical properties, and grammatical rules themselves.

Disjunctive type hierarchies

A fundamental type of knowledge organization are formal ontologies, which can be represented as directed acyclic graphs, or disjunctive trees. At the top, the most general type is found, and every type that stands below it is an subtype of this general type. The notion of "belowness" here is defined by an unbroken descending chain of edges from the supertype to the subtype. What it means for such a hierarchy to feature *inheritance* is that every property of any supertype, without exception, must also be true of the subtype. This information is snowballed down until one reaches the final subtype whose properties we want to know. This allows to eliminate redundancies, which can be inferred provided we know all supertypes of a given type.

A very simple example of such an ontology might be a dichotomy between regular and irregular phenomena; further down, one can divide regular phenomena into several different types of regular ones, and likewise irregular phenomena into different subtypes. I will use the example of English verbs: regular and irregular verbs are found, but cannot be both at the same time. Within regular verbs, one can distinguished those that end in a coronal stop and form their past tense with /ɪd/; those that end in a voiceless phoneme which is not a coronal stop and attach /t/ in the past tense; and the others whose past has a /d/. The kind of hierarchy I just described informally is crucially *disjunctive*: when a type has several daughter subtypes, any object which is an instance of the higher type must also be an instance of *some* daughter. Any regular verb forms its past with one of /ɪd, t, d/. What is more, this disjunction is by default *exclusive*: no objects can belong to two sister types at the same time. No verb can attach either /ɪd/ or /t/ in the past tense. Furthermore, if we call "leaves" those types which do not have any subtype, then the leaves form a partition of the root type of the tree: any member of the root type must be an instance of *some* leaf type. Furthermore, it can only ever be an instance of *a single* leaf type, never two or more: the leaf types are exclusive. This is exactly what this means for the hierarchy to be disjunctive and exclusive by default.

The improvement that a logic of typed feature structures as embraced by HPSG provides to ontologies is the shift from acyclic graphs to disjunctive semi-lattices: those relax the requirement that there be a single edge upwards from any point in the tree. This means that two daughters need not be mutually exclusive anymore: they may have a common subtype. A refinement of the ontology of English irregular verbs would allow the existence of verbs which can be both regular and irregular, like *to sneak* or *to hang*. Inheritance of information can be performed in the very same way as before, combining information from all supertypes of a given type. The difference is now that the information may not snowball down in one path, but there may be different sister supertypes contributing to the information of the subtype. One crucial fact about

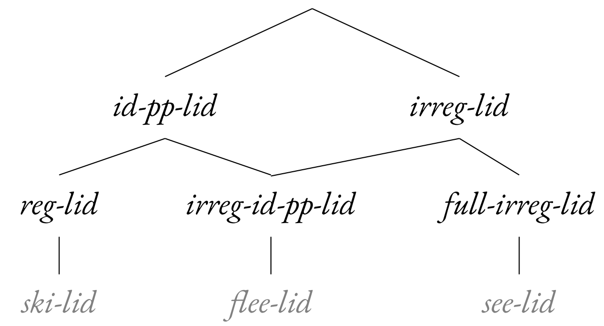

Figure 4.1: Type hierarchy with two supertypes, with a few elements in common. The term *lid* means "lexical identity".

common subtypes, however, is that two sister subtypes are interpreted as disjoint unless that is impossible. Thus, a disjunctive type hierarchy is exclusive by default, and may only become inclusive when the combination of two sister types is specifically licensed by the presence of a common subtype.

A refined example is found in figure 4.1. I split English verbs into three classes: all regular verbs, which have an identical form for past tense and past participle (*reg-lid*); irregular ones whose past tense and past participle are nevertheless identical (*irreg-id-pp-lid*); and those whose past forms are irregular, and different to one another (*full-irreg-lid*). Therefore, we can abstract two supertypes: the type of verbs with identical past and past participle; and the type of irregular verbs. Because they are prelinked to a common subtype, we are forced to tolerate elements common to those two supertypes. Inheritance in this hierarchy follows directly from the definition of the types, and is performed easily by combining information. When we know how to combine information, every information about a type can be inferred from where it stands in the hierarchy; i.e. what supertypes it shares an edge with. Note that even in a hierarchy which is not a tree, there are still so called leaf types, and these leaf types still partition the root type, because by definition they cannot have any common subtypes.

We can use an informal notion of information-combining for now, but a formal operation exists on typed feature structures which captures this notion precisely: that is unification. Because typed feature structures describe requirements on certain classes of objects, unification of two typed feature structures can often be performed by conjoining the requirements in both feature structures. This is sometimes also called intersection. To see why, call $D(\sigma)$ the set of objects described by σ, a typed feature structure; i.e. all those objects compatible with the requirements/properties that σ imposes. Then, the unification of σ with τ, notated $\sigma \sqcup \tau$, denotes exactly those objects which satisfy both the requirements of σ and the requirements of τ; but an object compatible with σ is precisely a member of $D(\sigma)$. Therefore, a member of $D(\sigma \sqcup \tau)$ must be a member of both $D(\sigma)$ and $D(\tau)$: this means exactly that $D(\sigma \sqcup \tau)$ is the intersection of $D(\sigma)$ and $D(\tau)$, i.e. $D(\sigma \sqcup \tau) = D(\sigma) \cap D(\tau)$. Therefore, unification seen as combining information, or requirements, corresponds exactly to intersecting classes of objects.

Online Type hierarchy: conjunctive hierarchies

With these notions of classification and inheritance in mind, let us turn to a device which significantly improves the abstractive powers of our classifications. Consider again the example in Figure 4.1. We may now want to capture the fact that the three lexemes at the bottom of the hierarchy all end in the vowel /i:/ in their base form, independently of their belonging to separate inflectional classes.

We now need a new supertype, which will be able to share elements with the three bottom classes, without encompassing them entirely (because each inflectional class also has lexemes that do not end in /i:/). Notice that this can be done, in a so-called "static" way: simply add a supertype next to the two that already exist, and for each class at the bottom, draw a new class which will be its intersection with the new type *ee-lid*, whose role is to constrain the final segment of its members. Then, each of the three lexemes at the bottom will belong both to its own inflectional class and to the type of lexemes that end in /i:/.

However, this already doubles the number of links we used to have in the hierarchy in Figure 4.1. The crucial idea behind Online Type hierarchies (Koenig & Jurafsky 1995, Koenig 1999) is that in addition to the distinction that Figure 4.1 draws between different inflectional classes, we can add *orthogonal* distinctions. What it means for these distinctions to be orthogonal is that the two dimensions of variation are largely independent: for instance, ending in /i:/ or not is independent from inflectional class. Conversely, knowing a verb's inflectional class does not allow to predict its final segment in any meaningful way. That is, subtypes in one dimension can combine freely, in principle, with any of the subtypes in the other dimension. By contrast, this is called "dynamic" cross-classification.

For our example, let us call $\boxed{\text{PAST}}$ the dimension which distinguishes different inflectional patterns; and $\boxed{\text{FINAL}}$ the one which distinguishes different finals for the verbs. The way that dimensions are drawn is in small caps, with a surrounding rectangle. This is represented in Figure 4.2. Now, each possible lexeme will have to inherit both an inflectional class and a final segment. For instance *ski*, *flee*, *see* each exemplify one inflectional class, and belong simultaneously to the *ee-lid* class. Likewise for *grieve*, *leave* and *weave*, except that they now belong to another type in the $\boxed{\text{FINAL}}$ dimension, namely the *eave-lid* type of verbs ending in /i:v/. This exemplifies the free combination which is possible between any type on the left-hand dimension, with any type on the right-hand dimension. This does not mean that all combinations will be meaningful: for instance, if no irregular verb besides *do* ends in /u:/, this means that the intersection of *irreg-id-pp-lid* and *oo-lid* is empty, and no verb can be a member of both classes. This

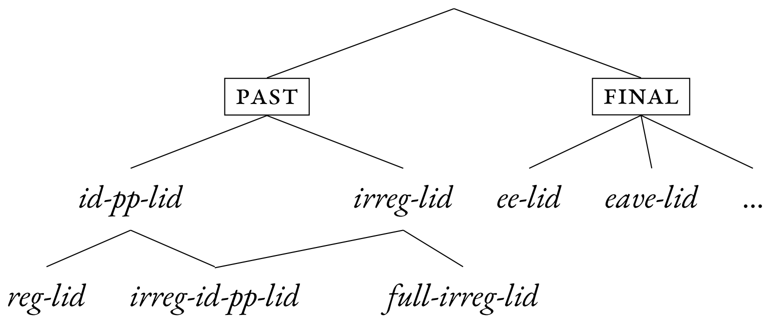

Figure 4.2: Example of an Online Type hierarchy: the largely independent dimensions of inflection and of final segment have been separated.

is an example of de-facto impossibility, but de-jure impossibility also exists, if two leaf types specify pieces of information which are in principle incompatible.

Now, notice that the leaf types, like *ee-lid* and *reg-lid* here, do *not* form a partition of the root type anymore. Indeed, *ski* is a lexeme which belongs to both those leaf types. This means that we now end up with hierarchies that are not fully *disjunctive* anymore. Instead, the dimensions are in a *conjunctive* relation to one another. While before, any element of the root type needed to belong to *some* daughter (mathematical disjunction/existential quantification), now any element of the root type must belong to *all* dimensions (mathematical conjunction/universal quantification). More precisely, for each relevant dimension, an element must belong to some daughter of that dimension. We can interpret this difference of behavior by saying that types behave *conjunctively* across dimensions, while they behave *disjunctively* within dimensions.

We can now rephrase the partitioning property that we had for disjunctive type hierarchies. While simple leaf types used to partition the root type, in an Online Type hierarchy, the actual leaf types are conjunctions of *pairs* of leaf types from each dimension. This means that leaf types can be multiplied across dimensions, in principle.

One last special case is allowed. This is that leaves are also allowed to belong to the two dimensions at the same time. For instance, we could want to specify that all inflectional classes seen thus far (*reg-lid*, *irreg-id-pp-lid* and *full-irreg-lid*) require that at least the first segment be identical between all forms (which is the case for virtually all verbs). Then, cases of suppletion do not fall into any of the neatly general classes of the PAST dimension: verbs like *be* and *go* can then directly be linked to the PAST dimension, without requiring that they belong to any of the previously established classes. While it does not find any supertype from that dimension, it can still perfectly belong to the *ee-lid* type in the FINAL dimension. This will be notated as in Figure 4.3. This possibility enables us to capture exceptional cases within a largely systematic alternation: those leaves will indeed be legitimate leaf types after cross-classification, since they are tied to both dimensions already. Such a leaf type is an exceptional case which is con-

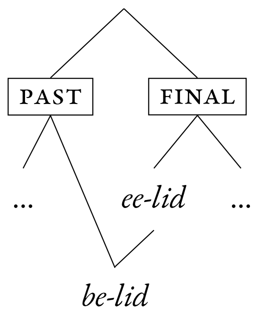

Figure 4.3: Example of prelinking with the verb *be*, which is suppletive but still ends in /i:/.

strained in both dimensions together instead of independently.

Let me summarize what we have seen for Online Type hierarchies. They feature organizing dimensions, and these dimensions tell us which types of objects are in principle compatible. The final leaf types of our hierarchy are not the leaf types of the dimensions, but are obtained by conjunctively combining, from each dimension, exactly one leaf type from that dimension. This enables class abstraction of systematic alternations between two dimensions, which thus eliminates redundancy. Information about the leaf rule types can then be recovered by cross-classification. In a sense, this is a limited version of open-world reasoning. Nevertheless, thanks to the safeguards I mentioned, we do not lose the ability to recognize exceptions even with open-world reasoning: we can tell when two options are in fact incompatible, and when special cases do not participate in the alternation. These statically encoded special cases can also allow to encode partial generalizations, or intermediate knowledge, in case an alternation between different types is not quite systematic.

4.1.2 Assumptions and principles of IbM

Let us now turn to specific facts about IbM in itself, beyond its recognizing Online Type hierarchies as its primary form of knowledge organization.

Assumptions

The fundamental object of IbM is not the morpheme, but the *realization rule*, in keeping with an inferential–realizational theory of morphology. These rules are recognized as ways of introducing *morphs*: these are really the equivalents of exponents or "rule effect" (e.g. string concatenation with a particular concatenative exponent, ablaut...) in PFM (Stump 2001). Critically, like in PFM, rules are not constrained to be 1 : 1 relations between function and form. They can be as general as one desires: this means recognizing $m : n$ relations as a fundamental possibility for inflection. More specifically, each rule type describes a list of the exponents that it introduces as a whole (MPH; morphs), and a set of the exponenda that the rule as a whole expresses (MUD; morphology under discussion). For instance, a rule such as that introducing the *kɛ-* prefix of Limbu, which serves to express second person, will have the description in (1). Here, it so happens that this rule is monomorphic, with a $1 : 1$ relation between function and form.

$$(1) \quad \begin{bmatrix} \text{MPH} & \left\langle \begin{bmatrix} \text{PH} & \text{kɛ} \\ \text{PC} & -\mathit{4} \end{bmatrix} \right\rangle \\ \text{MUD} & \left\{ \begin{bmatrix} \text{PER} & \mathit{2} \end{bmatrix} \right\} \end{bmatrix}$$

As we can see in (1), every exponent carries two pieces of information about itself: its phonology (PH) and its position (PC; position class). This incorporates a neo-templatic view of inflection where, in the limiting case, each exponent has a definitive slot. Making phonology and position attributes allows the same underspecification to occur for certain morphs, implementing variable morphotactics like free positioning of a morph, for instance (Crysmann & Bonami 2016). Naturally, defining the position classes and phonology–morphology interface is up to the analyst.

The morphosyntactic set will be the value of the attribute MS in any given word. It is a set (or, informally, an unordered bag) where every piece of relevant morphosyntactic information is found: lexical identity and stem allomorphs of the lexeme; syntactic–semantic information that the word carries in a specific instance of use, etc. This information is encapsulated into different set members within the MS set, which do not interact further. Since the set members describe objects, they can also feature attribute–value pairs within them, recursively. For instance, number and/or person information about a given participant, like $\begin{bmatrix} \text{PER} & \mathit{2} \end{bmatrix}$ in rule (1) above. Again, the syntax–morphology interface, i.e. the makeup of this set and its elements, and how information is split in different members, is up to the linguist.

Furthermore, in cases of overlapping exponence, IbM allows conditioning on any morphosyntactic context that can be described in the logic. This means that rules will also have an attribute MS (morphosyntax). This is where the conditioning context for the rule will be described. This introduces a content vs. context distinction in the form of MUD vs. MS. As Crysmann (2017) argues, the purported arbitrariness of such a distinction can often be avoided by resorting to Pure Sensitivity (Carstairs 1987).

Now that the building blocks of the theory have been explained, I will explain the principles which ensure the good use of those building blocks.

Principles

First, as I anticipated in the previous section, each word will relate a phonology, which should be the concatenation of a list of morphs, with a set of realization rules which are the ones that determine its morphology, and with a morphosyntactic property set. These facts are described in (2).

(2) $$\textit{word} \Longrightarrow \begin{bmatrix} \text{MPHS} & \textit{list(morph)} \\ \text{RR} & \textit{set(realisation-rule)} \\ \text{MS} & \textit{set(msp)} \end{bmatrix}$$

Then, rules should be able to condition on the content of the whole word. In particular, because the morphology they contribute should be part of the morphology of the whole word, the MUD set of any rule should be a subset of its MS set. This is expressed via tag identity in (3). The rest of the information there simply recapitulates the definition of a rule in the previous section.

(3) $$\textit{realisation-rule} \Longrightarrow \begin{bmatrix} \text{MUD} & \boxed{1}\,\textit{set(msp)} \\ \text{MS} & \boxed{1} \cup \textit{set(msp)} \\ \text{MPH} & \textit{list(morph)} \end{bmatrix}$$

Now that these technical requirements are out of the way, we need, like any morphological theory, a way to ensure that the recipes, i.e. the rules, available are actually applied when they are licensed. This is expressed by the requirement that all the rules' MUD sets together should equal the word's MS set. This is what the last line and the $\boxed{m_i}$ tags in (4) express. Thus, every exponenda in MS should be found within at least one rule of the RR set. This is enforces the principle of **morphological completeness**, which other theories use various devices to achieve; for instance, PFM (Stump 2001) implements it by the requirement to pass through every rule block.

(4) Morphological well-formedness (Crysmann 2023):

$$\textit{word} \Longrightarrow \begin{bmatrix} \text{MPH} & \boxed{e_1} \bigcirc \dots \bigcirc \boxed{e_n} \\ \text{RR} & \left\{ \begin{bmatrix} \text{MPH} & \boxed{e_1} \\ \text{MUD} & \boxed{m_1} \\ \text{MS} & \boxed{0} \end{bmatrix}, \dots, \begin{bmatrix} \text{MPH} & \boxed{e_n} \\ \text{MUD} & \boxed{m_n} \\ \text{MS} & \boxed{0} \end{bmatrix} \right\} \\ \text{MS} & \boxed{0} \left(\boxed{m_1} \cup \dots \cup \boxed{m_n} \right) \end{bmatrix}$$

All other information in (4) above, as well as other principles which are presented in e.g. Crysmann (2021b), simply enforce the intuition that the terminology suggests: because a rule can be conditioned on the morphosyntax of the whole word, it should be able to see it (tag identity of $\boxed{0}$'s); every exponent in a word should be licensed by some rule ($\boxed{e_i}$ tags), and they should all be combined in order of position classes, with sharing of position class prohibited.

The last principle that should be enforced is that rules should not overapply: this is **morphological coherence**. Crysmann (2023) suggests a restriction in rule combination which checks that two different rules do not have mutually subsuming descriptions. If that was not the case, they would violate coherence. This is the approach I take here. Note that this is different from most earlier versions of IbM (Crysmann &

Bonami 2016, 2017, Crysmann 2020, 2021a), in which the union in the MS set in (4) was required to be a disjoint union, where no two sets are allowed to overlap. Dropping this requirement allows for an economical treatment of dependent extended exponence: we can then safely assume that both rules with a dependent have a MUD element in common, as long as they do not mutually subsume each other.[1]

Two final grammar-writing tools are used. First, Pāṇini's principle, which will be developed in Section 4.2. However, one crucial idea to keep in mind is that this principle of specificity will only apply between *leaf rule types*. This is why identifying leaf types in Online Type hierarchy was such an important point in Section 4.1.1. We will come back to this.

Second, a device akin to PFM's "identity function default", which allows a rule to vacuously express any exponenda which does not find any corresponding rule, thus ensuring completeness despite this lack. This is especially relevant for cases of zero exponence.

4.1.3 Specific features of IbM

Just like the famous Paradigm Function Morphology of Stump (2001), IbM is an inferential–realizational theory of morphology. In the same way, it uses rules as recipes which allow to infer the form of a word on the basis of its function; there is no limit on the number of exponents that a rule introduces (n), nor to the number of elements of function it requires (m). The similarities stop there, however.

For one thing, IbM adopts a neo-templatic view, where position classes are possible properties of morphs, but properties which need not tie them down; PFM obligatorily assigns each rule to only one rule block. Moreover, the principle of competition in IbM applies entirely globally, and all possible pairs of leaf rule types are compared to ensure that neither is more specific than the other. This allows easy interpretation of phenomena like discontinuous bleeding (Crysmann & Bonami 2016). Like the rules, competition in PFM is specific to each rule block; this allows multiple exponence quite naturally whereas IbM needs additional assumptions like conditioning on MS instead of expression, to ensure that two rules with similar conditions both apply. This content/context distinction is one that PFM does not make.

Finally, the essential feature of IbM is its powerful ability to abstract over rules, which stems from the HPSG logic it is couched in. This is an abstractive power that PFM lacks, to the extent that rules which must be stipulated as disjunctions (this is done in Stump 2022). This enables generalizations beyond the partial generalizations over words that rules themselves constitute Crysmann & Bonami (2016), and allows leaf rule types to take care of the minute facts of competition and distribution, while

[1] Crysmann (2023) actually proposes another solution, where the union is non disjoint but simply requires mutual non-inclusion (so that each set contributes at least one unique element). There are computational reasons to disallow this kind of rule.

F \ G	−	+
α	**A**	**A**
β	B	C
γ	B	C

Table 4.1: Schematic pseudo-Pāṇinian split, repeated from Table 3.6b

$$\begin{bmatrix} \textit{A-rule} \\ \textsc{mph}\ \langle A \rangle \\ \textsc{mud}\ \{[\textsc{f}\ \alpha]\} \end{bmatrix}$$

(a) Cross-cutting override

$$\begin{bmatrix} \textit{B-rule} \\ \textsc{mph}\ \langle B \rangle \\ \textsc{mud}\ \{[\textsc{g}\ -]\} \end{bmatrix}$$

(b) One leg of the natural +/− split

$$\begin{bmatrix} \textit{C-rule} \\ \textsc{mph}\ \langle C \rangle \\ \textsc{mud}\ \{[\textsc{g}\ +]\} \end{bmatrix}$$

(c) The other leg of the split

Figure 4.4: Setup for pseudo-Pāṇinian competition

abstract rule types reify higher-level generalizations that can be made. This abstractive power, and the relative generality of the principles presented in Section 4.1.2, mean that IbM leaves a surprising lot to the analyst. Every single fact in a morphological analysis will be determined by the version of the principles assumed, the makeup of the morphosyntactic interface, and the ontological hierarchy of rules.

The next sections demonstrate how this formalism can be put to use to analyze the numerous puzzles of Limbu conjugation, presented in Section 3.2 and 3.4.

4.2 Approaching pseudo-Pāṇinian splits in IbM

Recall from Section 3.6a the characteristic pattern of pseudo-Pāṇinian splits: a natural split runs along some part of the paradigm, let's say in feature G, but the distinction is lost in a specific value of another feature, say the value α of feature F. This gives rise to the schematic representation in Table 4.1; at the level of rules, this translates to a logical situation like the descriptions in Figure 4.4. The rule introducing the exponent *A* can express any number of specific feature–value pairs in MUD – here, I choose to have only one.[2] Importantly, it should not specify the feature G; conversely, rules *B* and *C* should not be more specific than *A* about the features it expresses – here, F.[3]

[2] If needed, substitute $[\textsc{mud}\ \{[\textsc{f}\ \alpha]\}]$ for $[\textsc{mud}\ \{[\textsc{f}_1\ \alpha_1],[\textsc{f}_2\ \alpha_2],\ldots,[\textsc{f}_n\ \alpha_n]\}]$.

[3] In fact, all three rules could be restricted to the same part of the paradigm if they are all exponents of the same property $[\textsc{h}\ h]$; in the case of *-mʔna*, this would be the $[\textsc{num}\ pl]$ feature. I omit this here for the sake of clarity. The only thing that matters is that rule *A* is incommensurate with rules *B* and *C* on the whole, yet overrides both.

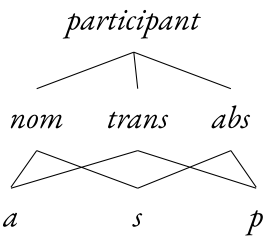

Figure 4.5: Type hierarchy for participant roles

Let us consider what a configuration such as this one would predict for the portmanteau *-mʔna* of Section 3.2.1. Given the distributions of exponents seen in Table 3.6a, and incorporating additional information about these markers' distributions, we can propose the tentative rule descriptions in (5). Anticipating the analysis in Section 4.4, I shall assume that *-u* and *-m* form a single complex rule. The morphosyntactic objects that stand for participants, such as $\begin{bmatrix} \textit{p} & \\ \text{PER} & \textit{3} \end{bmatrix}$, are organized in the type hierarchy given in Figure 4.5. The underspecified types that each combine two leaves together are given linguistically meaningful labels: *abs*, *nom*, *trans* (for "absolutive", "nominative" and "transitive"). The participant objects are themselves described using a person value. I assume the type hierarchy for person values given in Figure 4.6 (*adrs* stands for "addressee"). Similarly, Figure 4.7 presents the different types that can serve as values for number.

(5) a. $$\begin{bmatrix} \text{MPH} & \left\langle \begin{bmatrix} \text{PH} & \text{mʔna} \\ \text{PC} & \textit{2..5} \end{bmatrix} \right\rangle \\ \text{MUD} & \left\{ \begin{bmatrix} \textit{nom} & \\ \text{NUM} & \textit{pl} \\ \text{PER} & \textit{1excl} \end{bmatrix}, \textit{pret} \right\} \end{bmatrix}$$ Override

b. $$\begin{bmatrix} \text{MPH} & \left\langle \begin{bmatrix} \text{PH} & \text{i} \\ \text{PC} & \textit{6} \end{bmatrix} \right\rangle \\ \text{MUD} & \left\{ \begin{bmatrix} \textit{abs} & \\ \text{NUM} & \textit{pl} \\ \text{PER} & \textit{1}\vee\textit{2} \end{bmatrix} \right\} \end{bmatrix} \quad \begin{bmatrix} \text{MPH} & \left\langle \begin{bmatrix} \text{PH} & \text{u} \\ \text{PC} & \textit{4} \end{bmatrix}, \begin{bmatrix} \text{PH} & \text{m} \\ \text{PC} & \textit{5} \end{bmatrix} \right\rangle \\ \text{MUD} & \left\{ \begin{bmatrix} \textit{a} & \\ \text{NUM} & \textit{pl} \\ \text{PER} & \textit{1}\vee\textit{2} \end{bmatrix}, \begin{bmatrix} \textit{p} & \\ \text{PER} & \textit{3} \end{bmatrix} \right\} \end{bmatrix}$$

Competitors involved in the split

c. $$\begin{bmatrix} \text{MPH} & \left\langle \begin{bmatrix} \text{PH} & \text{ge} \\ \text{PC} & \textit{8} \end{bmatrix} \right\rangle \\ \text{MUD} & \left\{ \begin{bmatrix} \text{PER} & \textit{1excl} \end{bmatrix} \right\} \end{bmatrix} \quad \begin{bmatrix} \text{MPH} & \left\langle \begin{bmatrix} \text{PH} & \varepsilon \\ \text{PC} & \textit{2} \end{bmatrix} \right\rangle \\ \text{MUD} & \left\{ \textit{pret} \right\} \end{bmatrix}$$ Other competitors

Notice that, intuitively, the rule for cumulative *-mʔna* in (5a) has more information than any other: it has specific values of number, person and tense. However, the notion of specificity that IbM resorts to is more precise, and defined using rigorous logic.

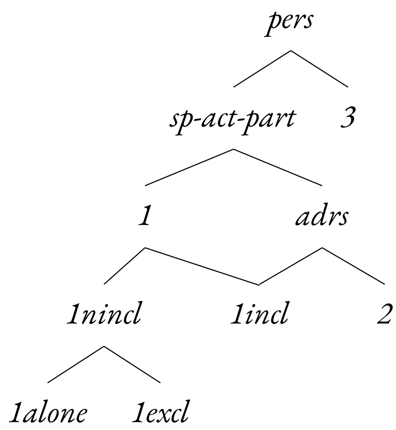

Figure 4.6: Type hierarchy of person values. See a more detailed model, where types are described by attribute–value pairs, in Appendix D.

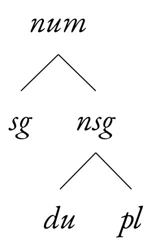

Figure 4.7: Type hierarchy for number values

Specifically, the notion of *subsumption* between two feature structure descriptions is a measure for informativeness. In this thesis, I define it as follows (but cf. Pollard & Sag 1987: Ch. 2 and Carpenter 1992):

Definition 1. Given two feature structure descriptions s and t, say that s subsumes t, and write $s \sqsubseteq t$, iff all of the following are true:[4]

- The type of s is a supertype of the type of t.
- For each attribute–value pair $\langle A, v_s \rangle$ in s, there is a pair $\langle A, v_t \rangle$ in t, such that
  - the attributes A are identical;
  - the value v_s subsumes the value v_t: $v_s \sqsubseteq v_t$.
- For each formula in s (i.e., path equation requiring structure sharing, or relation, or first-order-logic combination of these), the same formula holds in t.

Interpret any subsumption relation between lower-level feature structures (v_s, v_t, etc.) recursively. Write $s \sqsubset t$ iff $s \sqsubseteq t$ and $t \not\sqsubseteq s$.

Intuitively, if s subsumes t, then s is less informative, or less specific, or more general, than t. Each point in the definition above ensures that no information in s is lost in t; on the other hand t might have much more information than s. This is similar to set inclusion: every "information member" of s is also an "information member" of t. Subsumption is more powerful, however, because it is computed recursively and pieces of information can change (and become more specific) when going to a more specific feature structure; contrast this with the immutability of (atomic) set members.

Given this formal definition of subsumption, we can now formulate the conditions for Pāṇini's principle in IbM. Specifically, the cases in which it applies are defined via subsumption, while the operation that it effects simply requires first-order logic (Crysmann 2021b).

(6) **Pāṇinian Competition**

a. For any leaf type $t_1\begin{bmatrix}\text{MUD} & \mu_1 \\ \text{MS} & \sigma\end{bmatrix}$, say that $t_2\begin{bmatrix}\text{MUD} & \mu_2 \\ \text{MS} & \sigma \wedge \tau\end{bmatrix}$ is a morphological competitor, iff $\mu_1 \cup set \sqsubseteq \mu_2 \cup set$.

b. For any leaf type t_1 with competitor t_2, expand t_1's MS σ with the negation of t_2's MS $\sigma \wedge \tau$: i.e. $\sigma \wedge \neg(\sigma \wedge \tau)$ which is equivalent to $\sigma \wedge \neg\tau$.

The subsumption condition in (6a) deserves some comment. Informally, it enforces two things: (i) any element which is a member of μ_1 must have a corresponding member which is more specific in μ_2; and (ii) any set members which are specified as

[4] Richter (2021) uses $\sqsubseteq$ the other way around, as a "subtype" relation. Thus if $s \sqsubseteq t$, then s is more specific than t in his convention.

different in μ_1 must have different corresponding elements in μ_2. The union with an object of type *set* makes it possible to perform subsumption checks on sets of different cardinality.[5]

Let us see the effect of this operation on our rule types from (5). To check if two markers compete, we need not verify anything about the MS sets, because when there is no conditioning on MS in either rule, the condition on MUD and MS are equivalent by virtue of (4). Applying this to the rule for *-ge* in (5c), we can check that

$$\left\{\begin{bmatrix}\textsc{per} & \mathit{1excl}\end{bmatrix}\right\} \cup \mathit{set} \sqsubset \left\{\begin{bmatrix}\mathit{nom} & \\ \textsc{num} & \mathit{pl} \\ \textsc{per} & \mathit{1excl}\end{bmatrix}, \mathit{pret}\right\} \cup \mathit{set}.$$

Indeed, the only member of the left-hand-side description, $\begin{bmatrix}\textsc{per} & \mathit{1excl}\end{bmatrix}$, has a corresponding element on the right, which it subsumes. The additional set member *pret* on the right does not hurt because of the union with a generic *set* on the left: evidently, $\{\mathit{pret}\} \cup \mathit{set}$ is more specific than *set*. The very same reasoning applies for the rule introducing *-ɛ*.

Therefore, the (5a) rule does qualify as a competitor to the rules in (5c). By Pāṇini's principle (6), the latter rules' MS descriptions are expanded so that neither rule can appear when the more specific *-mʔna* is available: the result is found in (7).

(7) a. $$\begin{bmatrix}\textsc{mph} & \left\langle\begin{bmatrix}\textsc{ph} & \text{ge} \\ \textsc{pc} & \mathit{8}\end{bmatrix}\right\rangle \\ \textsc{mud} & \boxed{1}\left\{\begin{bmatrix}\textsc{per} & \mathit{1excl}\end{bmatrix}\right\} \\ \textsc{ms} & \boxed{1} \cup \mathit{set} \wedge \neg\left\{\begin{bmatrix}\mathit{nom} & \\ \textsc{num} & \mathit{pl} \\ \textsc{per} & \mathit{1excl}\end{bmatrix}, \mathit{pret}, \ldots\right\}\end{bmatrix}$$

b. $$\begin{bmatrix}\textsc{mph} & \left\langle\begin{bmatrix}\textsc{ph} & \varepsilon \\ \textsc{pc} & \mathit{2}\end{bmatrix}\right\rangle \\ \textsc{mud} & \boxed{1}\{\mathit{pret}\} \\ \textsc{ms} & \boxed{1} \cup \mathit{set} \wedge \neg\left\{\begin{bmatrix}\mathit{nom} & \\ \textsc{num} & \mathit{pl} \\ \textsc{per} & \mathit{1excl}\end{bmatrix}, \ldots\right\}\end{bmatrix}$$

This makes the right predictions, because *-ɛ* becomes unavailable precisely when in addition to preterite tense, a 1PES/A participant is present (in particular, it is still available for 1DES/A, 2PLS/A and 1PEP participants). Likewise, *-ge* is unavailable whenever some such participant is present, *and* the tense is preterite. It turns out that situations with two first-person-exclusive participants are disallowed in Limbu (recall the remark

[5]This is exactly analogous to the injectivity condition formulated in Section 2.5.2 of Loreau Unger (2022). Instead of competing rules, it was meant to apply to the MUD set C (for "content") of a rule, compared to the MS set S (for "scenario") of the word.

on page 11), so it is exactly that participant whose exclusive person could be expressed by *-ge* which will cause the appearance of *-mʔna* instead.

On the other hand, consider the other rules expressing number, in (5b). In both cases due to the recursive nature of subsumption, we should check that: $pl \sqsubseteq pl$, which is trivially true, and $1{\vee}2 \sqsubset 1excl$, which is true given the appropriate type hierarchy in Figure 4.6, where $1{\vee}2$ is the same as *sp-act-part* (for "speech act participant"). With respect to both these constraints, the *-mʔna* rule is the more specific one. However, when comparing the types of participants, this specificity breaks down: for the *-u-m* rule, we have $a \not\sqsubseteq nom$ – in fact, the reverse relation is true because *a* is the subtype. For the *-i* rule, *abs* and *nom* are not even comparable for the subtype relation. What is more, the inclusion of MUD sets is not even possible for *-u-m*, as there is no [PER *3*] element in the MUD of the rule for *-mʔna*. Therefore, this violates the informal condition (i) that was formulated for subsumption on page 47).

If we take the descriptions in (5) at face value, the prediction of the formalism is that the rules are compatible in a given word, since none of the rules strictly subsumes the other. Cooccurrence, however, is not what we find, because *-mʔna* blocks appearance of both *-i* and *-u-m*, in their respective environments.

Pseudo-Pāṇinian splits present us with the dilemma that we want to generalize over the overrides, while at the same time we need their description to be specific enough to capture their blocking potential by Pāṇini's principle.[6] This problem is in no way specific to the formalism chosen here, but it will plague any theory of morphology that subscribes to a version of Pāṇini's principle in terms of specificity of rule description. With the exception of Anderson (1992)'s A-morphous Morphology, which uses extrinsic rule ordering instead, this includes pretty much every theory of morphology, including Lexical Phonology Morphology (Kiparsky 1985), lexical–inferential Distributed Morphology (Noyer 1992), Paradigm Function Morphology (Stump 2001), or the defaults-based Network Morphology (Brown & Hippisley 2012).

In Section 3.2.1, I argued that *-mʔna* is best considered an override, because it can preempt as many as four markers in one fell swoop; moreover, competition is really enacted separately in transitive and intransitive lexemes, so that the distinction between the two areas delimited by the split is well motivated. This line of analysis can be implemented using the static type hierarchy of rules in Figure 4.8. The top rule abstracts four pieces of information common to both subtypes, and separates into one for each specific cell where *-mʔna* appears. Recall that because we are using inheritance of properties from supertypes to subtypes, any property of form or function specified at the top is automatically inferred in the two bottom leaves. Therefore, redundant information need not be repeated.

[6] This has not been noted before: van Driem (1987) does not provide a formal analysis (quite aptly, listing exceptions wherever they appear), and Stump (2022)'s fragment of Limbu does not include the intransitive paradigm.

$$\begin{bmatrix} \text{MPH} & \left\langle \begin{bmatrix} \text{PH} & \text{mʔna} \\ \text{PC} & \textit{2..5} \end{bmatrix} \right\rangle \\ \text{MUD} & \left\{ \begin{bmatrix} \textit{nom} & \\ \text{PER} & \textit{1excl} \\ \text{NUM} & \textit{pl} \end{bmatrix}, \textit{pret}, \ldots \right\} \end{bmatrix}$$

$$\begin{bmatrix} \text{MUD} & \{s, [\,]\} \end{bmatrix} \qquad \begin{bmatrix} \text{MUD} & \left\{ a, [\,], \begin{bmatrix} p & \\ \text{PER} & 3 \end{bmatrix} \right\} \end{bmatrix}$$

Figure 4.8: Rule hierarchy describing two rules involving *-mʔna*

This radically changes the way that Pāṇinian competition is resolved, because there are now two specific leaf types instead of a general one as in (5a). The competition is then tested separately for each leaf type, and this renders the intransitive variant (left) to preempt the *-i* rule in (5b); while the transitive variant (right) can outcompete *-u-m*. Indeed now we find:

Competition between intransitive *-mʔna* (Figure 4.8) and *-i* (5b)

$$\left\{ \begin{bmatrix} \textit{abs} & \\ \text{NUM} & \textit{pl} \\ \text{PER} & \textit{1} \lor \textit{2} \end{bmatrix} \right\} \cup \textit{set} \sqsubset \left\{ \begin{bmatrix} \textit{s} & \\ \text{NUM} & \textit{pl} \\ \text{PER} & \textit{1excl} \end{bmatrix}, \textit{pret} \right\} \cup \textit{set},$$

because $\textit{abs} \sqsubset s$ (unlike $\textit{abs} \not\sqsubseteq \textit{nom}$ earlier), and likewise

Competition between transitive *-mʔna* (Figure 4.8) and *-u-m* (5b)

$$\left\{ \begin{bmatrix} \textit{a} & \\ \text{NUM} & \textit{pl} \\ \text{PER} & \textit{1} \lor \textit{2} \end{bmatrix}, \begin{bmatrix} p & \\ \text{PER} & 3 \end{bmatrix} \right\} \cup \textit{set} \sqsubset \left\{ \begin{bmatrix} \textit{a} & \\ \text{NUM} & \textit{pl} \\ \text{PER} & \textit{1excl} \end{bmatrix}, \textit{pret}, \begin{bmatrix} p & \\ \text{PER} & 3 \end{bmatrix} \right\} \cup \textit{set}.$$

With this, we can now expand each rule into its version which is preempted when the appropriate version of *-mʔna* appears instead:

(8) a.
$$\begin{bmatrix} \text{MPH} & \left\langle \begin{bmatrix} \text{PH} & \text{i} \\ \text{PC} & \textit{6} \end{bmatrix} \right\rangle \\ \text{MUD} & \boxed{1} \left\{ \begin{bmatrix} \textit{abs} & \\ \text{NUM} & \textit{pl} \\ \text{PER} & \textit{1} \lor \textit{2} \end{bmatrix} \right\} \\ \text{MS} & \boxed{1} \cup \textit{set} \land \neg \left\{ \begin{bmatrix} \textit{s} & \\ \text{NUM} & \textit{pl} \\ \text{PER} & \textit{1excl} \end{bmatrix}, \textit{pret}, \ldots \right\} \end{bmatrix}$$

b.
$$\begin{bmatrix} \textsc{mph} & \left\langle \begin{bmatrix} \textsc{ph} & \mathrm{u} \\ \textsc{pc} & \mathit{4} \end{bmatrix}, \begin{bmatrix} \textsc{ph} & \mathrm{m} \\ \textsc{pc} & \mathit{5} \end{bmatrix} \right\rangle \\ \textsc{mud} & \boxed{1} \left\{ \begin{bmatrix} \mathit{a} & \\ \textsc{num} & \mathit{pl} \\ \textsc{per} & \mathit{1 \vee 2} \end{bmatrix}, \begin{bmatrix} \mathit{p} & \\ \textsc{per} & \mathit{3} \end{bmatrix} \right\} \\ \textsc{ms} & \boxed{1} \cup \mathit{set} \wedge \neg \left\{ \begin{bmatrix} \mathit{a} & \\ \textsc{num} & \mathit{pl} \\ \textsc{per} & \mathit{1excl} \end{bmatrix}, \mathit{pret}, \ldots \right\} \end{bmatrix}$$

Finally, a similar behavior happens for the two rules *-ɛ* and *-ge*. I only write the result of competition for the former in (9), because the latter is analogous. It has not one but two competitors, which are each of the subtypes in Figure 4.8.

(9)
$$\begin{bmatrix} \textsc{mph} & \left\langle \begin{bmatrix} \textsc{ph} & \varepsilon \\ \textsc{pc} & \mathit{2} \end{bmatrix} \right\rangle \\ \textsc{mud} & \boxed{1} \{\mathit{pret}\} \\ \textsc{ms} & \boxed{1} \cup \mathit{set} \wedge \neg \left\{ \begin{bmatrix} \mathit{s} & \\ \textsc{num} & \mathit{pl} \\ \textsc{per} & \mathit{1excl} \end{bmatrix}, \ldots \right\} \wedge \neg \left\{ \begin{bmatrix} \mathit{a} & \\ \textsc{num} & \mathit{pl} \\ \textsc{per} & \mathit{1excl} \end{bmatrix}, \begin{bmatrix} \mathit{p} & \\ \textsc{per} & \mathit{3} \end{bmatrix}, \ldots \right\} \end{bmatrix}$$

In the end, this static generalization across rule types presented in Figure 4.8 does allow a unified treatment of pseudo-Pāṇinian splits as a special case of regular Pāṇinian competition. Thanks to inheritance, we can still capture the properties that are common to both instances of *-mʔna*. This twin-like behavior of the override can be explained by its already specific nature, which comparatively justifies idiosyncratic specifications.

4.3 Non-singular marking and *-si*

In Section 3.2.2, I argued for an analysis of *-si* as a general elsewhere marker of non-singular number. As I pointed out, the existence of at least two position classes for it leads us to posit two different markers: one for A functions and one for P functions. I choose to count S functions together with the A marker, leading to the two variants on the left of Figure 4.9.

4.3.1 Pseudo-Pāṇinian–like characteristics

Recall from Section 3.2.2 that *-si* has a very wide distribution across the paradigm, encoding non-singular whenever there is no plural competitor, that will coerce it to dual. One specific problem we identified concerned the relative distribution of *-si* and *mɛ-*, giving rise to a morphomic split in the shape of two interlocking L's. Let us therefore turn to the properties of *mɛ-*. It participates in a natural split in S and A>3 functions, but it is able to override the dual functions of *-si* in case the third-person participant is an A acting on a first or second person.

This is pseudo-Pāṇinian in two respects: on the one hand, the split is disrupted in the A>1/2 part of the paradigm, leading to a paradigm layout like the one in Table 3.7, which is similar to that of Table 3.6b. On the other hand, the override has a distribution which is hard to capture: the smallest natural description of the positions of *mɛ-* would be 3NSGA/S; but this coarse description predicts too many overrides, notably in the 3DUS cells. Contrast this with the situation with *-mʔna*, where the coarse description predicts too few overrides. Therefore, the analogy between this split and the previous one which I will pay attention to here is not one of general layout. Rather, it stems from the discrepancy between generalizations that can be made about the override and the actual distribution of the override.

The solution I propose is analogous to that for *-mʔna* in Figure 4.8: I generalize the common properties of every instance of the marker into an abstract rule type, but predict the specific distribution using more fine-grained leaf types. This is presented on the right of Figure 4.9.

Given these rule descriptions, the predicted distribution matches directly the observed distribution: no *mɛ-* is predicted in the 3NSGP cells, because the abstract *mɛ-* rule type is only nominative, i.e. S or A. Conversely, no *-si* is predicted in the 3NSG>1/2 scenarios because the *mɛ-* leaf type on the left is more specific than the expected "agent *-si*" rule. Finally, the naturalness of the split in the middle section of Table 3.7 is accounted for by another leaf type: *mɛ-* can simply take on a plural function in general. Note that this leads to a spurious ambiguity in 3PL>1/2 cells: either *mɛ-* rule is available then. I choose to ignore this ambiguity, because it is irreducible in the distribution of the marker, and it allows to keep natural generalizations in the analysis. Indeed, choosing to have a rule expressing 3DU>1/2 would seem to ignore the blatant fact of number neutralization in 3NSG>1/2 scenarios.

Now that the third-person number split has received an analysis, let me sketch how Pāṇinian competition with other plural markers allows *-si* to receive a dual function in many corners of the paradigm. The rules for markers that serve to express plural are given in Figure 4.10. The rules that are found there have much more information than the very generic non-singular *-si*, in either flavor. All serve to express at least a plural participant, with additional conditions on its person. Consider the example of the *-u-m* rule on the bottom left: it does not have any particular MS condition, so we can use the condition in (6a) while concentrating on the MUD descriptions. Again, we find that

$$\left\{\begin{bmatrix} \mathit{nom} \\ \textsc{num}\ \mathit{nsg} \end{bmatrix}\right\} \cup \mathit{set} \sqsubset \left\{\begin{bmatrix} \mathit{a} \\ \textsc{num}\ \mathit{pl} \\ \textsc{per}\ \ \mathit{1{\vee}2} \end{bmatrix}, \begin{bmatrix} \mathit{p} \\ \textsc{per}\ \mathit{3} \end{bmatrix}\right\} \cup \mathit{set},$$

so Pāṇini's principle should apply to this rule, and we can expand the description of the *-si* rule into the description in (10):

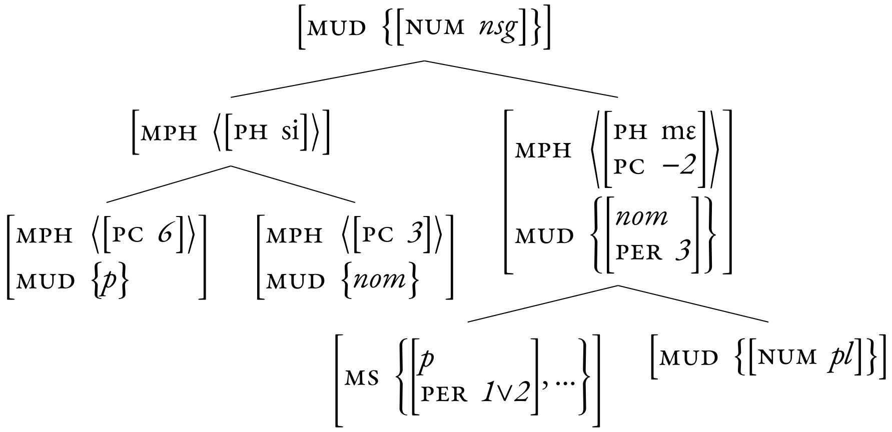

Figure 4.9: Rule hierarchy for third-person number markers

(10)
$$\begin{bmatrix} \text{MUD} & \boxed{1}\left\{\begin{bmatrix} nom \\ \text{NUM } nsg\end{bmatrix}\right\} \\ \text{MS} & \boxed{1} \cup set \wedge \neg\left\{\begin{bmatrix} a \\ \text{NUM } pl \\ \text{PER } 1\vee 2\end{bmatrix}, \begin{bmatrix} p \\ \text{PER } 3\end{bmatrix}, \ldots\right\}\end{bmatrix}$$

Therefore, whenever the scenario is (1∨2)A>3 and nominative *-si* appears, we know that it cannot be in a plural function: it becomes restricted to $nsg \wedge \neg pl$, i.e. dual (cf. Figure 4.7).

This shows that when the conditions for a plural marker are met, Pāṇinian competition will remove the cells from the possible distribution of the potential *-si* rule. In a contrasting dual cell, however, *-si* will be able to appear. Crucially, because no other plural or non-singular marker is available for agent first persons acting on second persons, we predict that only *-si* will be found in such cells. This is what happens with *-nɛ-tchi-ge* 1NSG>2. The entirety of the preceding discussion is faithful to the line of analysis I sketched at the end of Section 3.2.2.

Before we move on, let us fix one blind spot. We have eluded until now the conspicuous absence of an *-i* marker in certain 1PI scenarios. This can be seen in Tables 3.3 and 3.6a. Specifically, the 1PIS and 3>1PI cells do not receive any overt marking of number. This is surprising on two accounts. First, *-i* seems to be the absolutive counterpart of *-m*, available for any first or second person plural. Second, even granting that *-i* may not appear specifically in the inclusive, the paradigm could appeal to a general *-si* marker in this area, too, just like it does for other cells: 3NSGP and 1NSG>2. Of course, there is still an interest in keeping a number distinction in the inclusive, and it so happens that the absence of a number marker can fulfill this function. This is an argument in favor of a significant zero in Limbu, specifically in the 1PIS/P corner of the paradigm.

Because the full form is *a-* (which in other circumstances is a quite wide "2>1 or

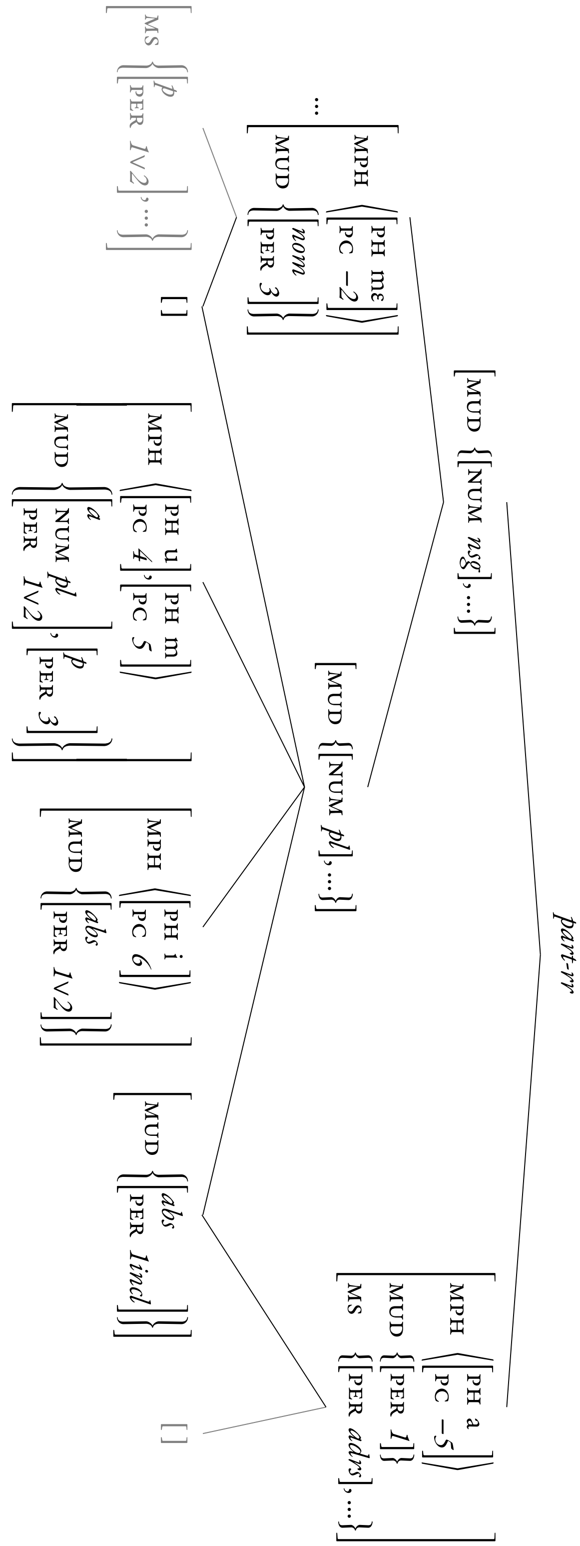

Figure 4.10: Rule hierarchy for plural markers

1INCL" prefix), there are in principle two ways of going about this. On the one hand, two rules could introduce *a-* as a first-inclusive marker, and no morphs as a plural marker, respectively; the latter rule would preempt appearance of *-i*. On the other, one single rule could license *a-*, expressing first-plural-inclusive all at once. However, in the former case, we should make sure that the two rules do not stand in a subsumption relation. The zero rule would need to be conditioned on the plural participant being first-person-inclusive; but this is already more specific than general *a-*.[7] Choosing a single *a-* rule, as a subtype of a more general *a-* rule, eschews such competition problems. The resulting analysis is found on the right-hand side of Figure 4.10. The more general rule would express first person, conditioned on presence of an addressee: this allows describing the "2>1 or 1INCL" distribution by a single non-disjunctive term.[8] With this "significant zero", we can explain both the absence of *-i* and the absence of *-si* in the 1PIS/P corner, because this very specific marker has the power to preempt both.

One last adjustment is needed for *-i*, however. I shall propose a temporary solution here, and then show in the next section how this issue can be dealt with in conjunction with the independent account of uninflectedness given there. For now, because $nom \nsubseteq abs$, it is not more specific than nominative *-si* in terms of role in the intransitive paradigm. So its rule should be split into two sub-rules: one for S functions (which will be more specific than nominative *-si*), and one for P functions (more specific than patient *-si*). This is a natural consequence of grouping S and A functions together for *-si*.

4.3.2 Uninflectedness

As we saw in Section 3.2.3, number marking is available, in some form, both in the intransitive paradigm and in the part of the transitive paradigm involving third person. More specifically: there is always some non-singular marker in those scenarios. This means that third-person non-singular participants always have an associated exponent. However, in the "local" part of the paradigm, where both arguments are speech act participants, number marking is far more restricted. In 1>2 scenarios, the number of the second-person patient is not marked except if the agent is singular; a single non-singular marker (*-si*) is available for person of the first-person agent. In the reverse 2>1 scenarios, no number is marked whatsoever. Interestingly, the exclusive marker *-ge* behaves just like a number marker for the purposes of its distribution: it is found exactly in S,

[7]This content/context distinction cannot operate at full power on identical members of the MS set: difference between $\begin{bmatrix} \text{MUD} & \boxed{1}\left\{\begin{bmatrix}\text{NUM} & pl\end{bmatrix}\right\} \\ \text{MS} & \boxed{1}\left\{\begin{bmatrix} abs & \\ \text{PER} & 1incl\end{bmatrix}\right\} \cup set \end{bmatrix}$ and $\begin{bmatrix} \text{MUD} & \boxed{1}\left\{\begin{bmatrix} abs & \\ \text{NUM} & pl \\ \text{PER} & 1incl\end{bmatrix}\right\} \\ \text{MS} & \boxed{1} \cup set \end{bmatrix}$?

[8]The 1>2 cells have their own portmanteau *-nε*, $\begin{bmatrix} \text{MUD} & \left\{\begin{bmatrix} a & \\ \text{PER} & 1\end{bmatrix}, \begin{bmatrix} p & \\ \text{PER} & 2\end{bmatrix}\right\}\end{bmatrix}$, which blocks *a-*.

EXCL<>3 and 1EXCL>2 scenarios.

I will ignore the 1SG>2 part of the paradigm, because any number marking there involves the dependent suffix *-ŋ* (cf. Section 3.4). For now, I consider that this area is not marked for number at all: I defer the analysis for the unexpected appearance of marking to Section 4.4. For the same reason, I omit a rule for *-u-m*, even though one appeared in Figure 4.10. Both of these cases will be (re)integrated later, in a part of the hierarchy specific to carrier-dependent phenomena (in Section 4.4).

We have seen in Section 3.2.3 that postulating zero overrides to explain the unavailability of certain markers does not seem a fruitful line of analysis. Instead, I suggest to identify zones of inflectedness, which have a reality independent of the specific exponents that they license. Specifically, three such areas emerge:

- the **intransitive paradigm,** which allows all the compatible number markers seen in Figure 4.10 as well as non-singular *-si* (specialized to dual): third plural *mɛ-*, first and second plural *-i*, inclusive plural *a-* and exclusive *-ge*;
- all parts of the transitive paradigm **with a third-person participant,** which draw out an L-shape on the bottom and right of Table 3.4 and receive patient plural *-i* and *a-*, patient non-singular or dual *-si*, agent third non-singular or plural *mɛ-*, agent plural *(-u)-m*, and exclusive *-ge*;
- the **1>2 square area,** only for first-person agents; licensing non-singular *-si* and exclusive *-ge*.

By contrast, none of the markers mentioned above appear in the other areas of the paradigm. Namely, no *-i*, *-m*, *-si* or even *-ge* are found in the reverse 2>1 square; and neither *-i* nor *-si* are found to express *patient* number (of the second person) in 1>2.

As we can see, the function of the markers is left almost intact across these different areas of inflectedness. Conversely, the areas of inflectedness are largely the same for each marker, independently of its function. This two-way systematicity strongly suggests making use of cross-classification between inflectedness conditions, on the one hand, and actual markers, on the other. This is represented in Figure 4.11. All the markers we have already seen in Figures 4.9 and 4.10, in addition to *-ge*, are now grouped in the $\boxed{\text{EXPO}}$ dimension ("exponence"), which faces the $\boxed{\text{INFL}}$ dimension ("inflectedness").[9] As we saw in Section 4.1.1, leaf rule types are then obligatorily formed by systematically combining leaf types from each dimension: the markers we provided rules for earlier must now be systematically combined into daughter types to yield the actual rules that will make them appear.

This enforces the appearance of each marker precisely in the areas where it has been noticed, and conversely, this implicitly bans it in areas outside its distribution. For in-

[9] I omit certain supertypes, like [MUD {[NUM *pl*]}], from Figure 4.10, for the sake of clarity. Similarly, the rule type for *a-* 1PI(S∨P) is recapitulated without some of its supertypes: I write the redundant information for the sake of completeness.

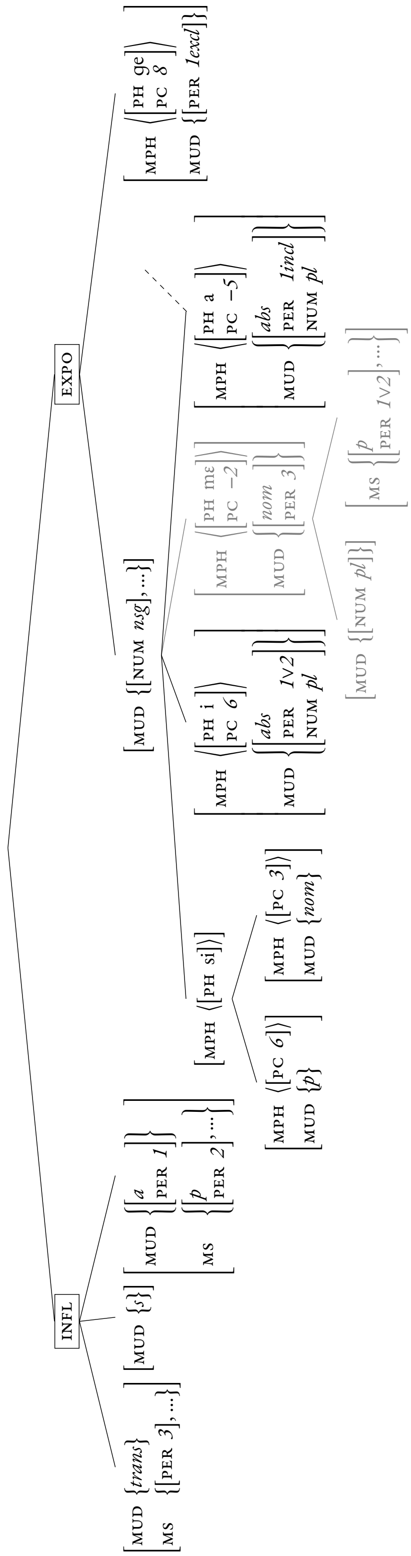

Figure 4.11: Rule hierarchy which constrains the appearance of non-singular markers

stance, while earlier rules allowed appearance of *-si* in 2>1 scenarios, this is now excluded. This is because no MS description in the three types of $\boxed{\text{INFL}}$ is compatible with a 2>1 scenario, thus excluding the appearance of any marker in $\boxed{\text{EXPO}}$ in this case.[10]

One last remark is in order concerning preemption of non-singular *-si* by absolutive plural *-i*. In Section 4.3.1, we had to stipulate specialization of *-i*, in order to make Pāṇinian competition work. However, given that we independently need to define zones of inflectedness, and that these zones entail a distinction between intransitive and (some) transitive scenarios, cross-classification of absolutive *-i* with transitivity will independently specialize the more general *abs* to either *s* or *p*. As a result, no stipulation for the speciifc case of *-i* is needed, but it rather follows from cross-classification with inflectedness constraints.

In a word, the analysis in Figure 4.11 allows to restrict the appearance of markers without changing their content (MUD) much, only their context (MS). This comes at the cost of up to three rule subtypes for each marker, but in the limiting case, some markers like patient *-si* only find one logically compatible $\boxed{\text{INFL}}$ leaf. Moreover, this breaking down of rules into particular subtypes is in line with the analysis of the previous two phenomena; what we gain here is also the fact that the subtypes can be described systematically by cross-classification, significantly saving on description length.

4.4 Carrier-dependent marking

Recall from Section 3.4 that the two morphs *-ŋ* and *-m* were morphotactically dependent on three different carriers morphs: *-u*, *-si* and *-i*. In addition to this dependence, multiple exponence is allowed, so that as many dependents can be found in the form as there are carriers: zero, one, or two. A highly similar behavior is found in Batsbi gender markers; when many carriers are found in a word, this gives rise to "exuberant exponence" (Harris 2009). These facts were analyzed in IbM by Crysmann (2021a). Here, I will show that the same arguments and formal mechanisms can be deployed for an analysis of Limbu, and I shall highlight additional advantages of this approach, namely in allowing the reverse situation where carriers are licensed by the dependent.

Carriers serve to express the same information with or without a dependent, and irrespective of the exact identity of the dependent. Conversely, dependents function in the same way regardless of the carrier they attach to. This is why the two markers can be so easily separated. Despite this independence, however, the morphotactic dependency needs to be captured: this can be done by means of composing the two parts into a single

[10]Notice that for third-person markers like *mɛ-*, the cross-classification with $\boxed{\text{INFL}}$ only separates transitive and intransitive cases. Indeed, unification with $\left[\text{MS}\ \{[\text{PER}\ \mathit{3}], \ldots\}\right]$ is trivial; and they cannot serve to express properties of a first person, so unification fails with the last inflectedness condition, $\left[\text{MUD}\ \{[\text{PER}\ \mathit{1}]\}\right]$.

complex rule. Akin to our treatment of inflectedness in Section 4.3.2 above, where we systematically combined constraints from two orthogonal dimensions, we shall agin rely on cross-classification by letting one dimension introduce the carrier, and leaving to another dimension the decision of whether to insert a dependent, and which one.

In the analysis I am about to propose, I will consider both inflectedness, i.e. the conditions under which individual markers can appear, and carrier-dependent marking as morphotactic in nature. I will therefore integrate the type hierarchy of dependent markers into the $\boxed{\text{INFL}}$ dimension of Figure 4.11, which I shall now call $\boxed{\text{MORPHOTAX}}$, in order to reflect the morphotactic nature of inflectedness and dependent exponence. Most importantly, we will complement the original inflectedness constraints in this dimension with constraints for dependent markers. The result of this is the hierarchy in Figure 4.12. In order to permit bimorphic rules in addition to the monomorphic rules we had until now, several minor adjustments are necessary, to which I shall turn now.

First, rules for dependent markers in the $\boxed{\text{MORPHOTAX}}$ dimension are bimorphic. In order to have them combine with rules in the $\boxed{\text{EXPO}}$ dimension, carrier markers must be allowed to be bimorphic as well. In the same way, the MUD description should not restrict the set to a single element, to leave space for the dependent to express its exponenda, if needed. These two problems can be resolved by simpling opening the list and set, respectively: instead of $\left[\text{MPH } \left\langle [\,] \right\rangle\right]$, write $\left[\text{MPH } \left\langle [\,] \right\rangle \oplus \textit{list}\right]$, or equivalently $\left[\text{MPH } \left\langle [\,], \ldots \right\rangle\right]$. Thus, the remainder of the MPH list can contain a dependent, or be empty if there is none, leaving a monomorphic rule as before. This is analogous for MUD, with sets instead of lists and set union instead of list concatenation. As for morphs which cannot be carriers at all, both values can be kept closed, as they do not allow any additional material inside their rules anyway. The rule for nominative *-si*, drawn in grey in Figure 4.12, provides an example of such a non-carrier rule type.

Second, we need to integrate the possibility of dependents into the dimension on the left, now called $\boxed{\text{MORPHOTAX}}$. Keeping the different rule types that specify inflectedness, we make them explicitly monomorphic to keep their cross-classified subtypes identical. This is by contrast to the dependent rules which will be bimorphic. Therefore, these types formerly in $\boxed{\text{INFL}}$ now become subtypes of the monomorphic rule type, within the $\boxed{\text{MORPHOTAX}}$ dimension: they are represented by the grey ellipsis in Figure 4.12. Then, we retrieve the former fully specified rule types by cross-classifying rules in $\boxed{\text{EXPO}}$ with leaf types within the monomorphic type. On the other hand, former "inflectedness" is not relevant for carrier-dependent rules in its current form; I will return to this later.

Let us turn to the elements which specifically take care of the different dependent rules. Choosing the right position class indices, we can capture the fact that dependents always follow their carriers, and are always phonologically nasals. Because carriers can occur in different positions, we should leave the position class for them open, but still capture the sequence by linking together the positions of both morphs. Carrier-

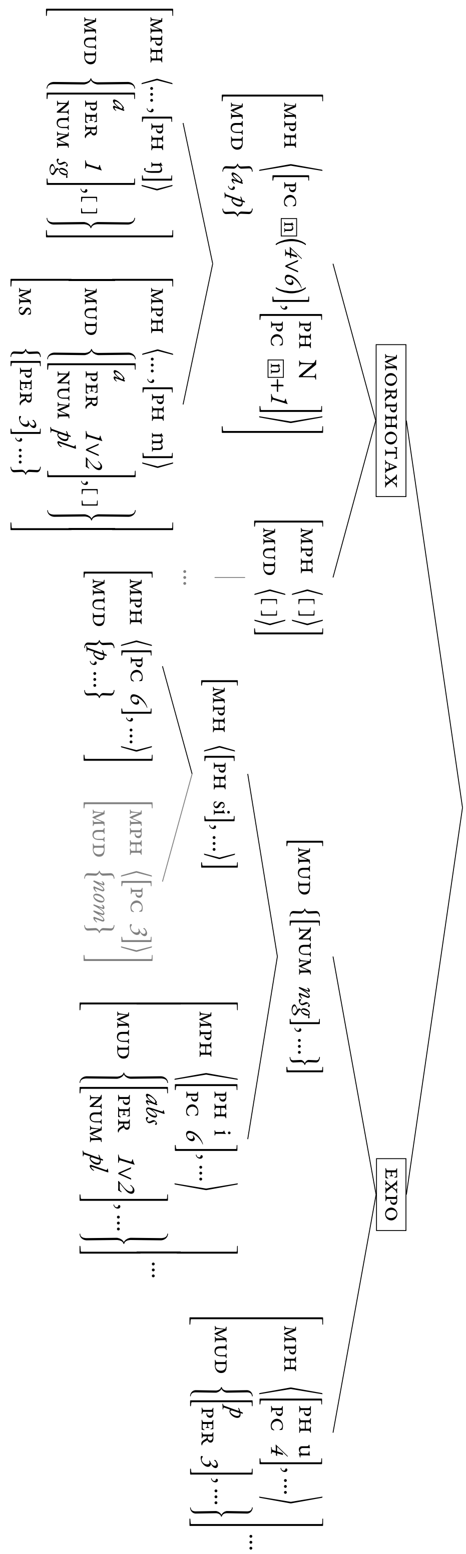

Figure 4.12: Rule hierarchy describing dependent–carrier rules

dependent rules encode information for both the agent and the patient, and this is reflected in the MUD set. Finally, the choice of the dependent is encoded by the distinction between the two subtypes of dependent rules: *-ŋ* expresses agent first-person singular, while *-m* expresses plural speech-act-participant agents. The fact that *-m* cannot express plural agents in local configurations (1>2 and 2>1) is captured by the conditioning on a third person in MS.

(11)
$$\begin{bmatrix} \text{MPH} & \left\langle \begin{bmatrix}\text{PH} & \text{u}\\ \text{PC} & 4\end{bmatrix}, \begin{bmatrix}\text{PH} & \text{ŋ}\\ \text{PC} & 5\end{bmatrix}\right\rangle \\ \text{MUD} & \left\{ \begin{bmatrix} a & \\ \text{PER} & 1 \\ \text{NUM} & sg\end{bmatrix}, \begin{bmatrix} p & \\ \text{PER} & 3\end{bmatrix}\right\} \end{bmatrix}$$

Cross-classification then gives us three rules with a dependent *-ŋ*, and three with a *-m* dependent. For instance, combining *-u* and *-ŋ* yields the leaf rule type in (11). As we have seen in Table 3.11, this deals with *-ŋ* dependents off the shelf. Two remarks are needed for *-m*, though: first, conditioning on a third person recapitulates the fact that *-si-m* does not always occur, notably not in local scenarios, noticed in Section 3.4. Second, we recognize *-i-m* as a rule type in this analysis, in contradiction to Table 3.11. However, the layout of the paradigm is such that one cannot have three participants to inflect for in the same word form. Therefore, we predict that this rule cannot surface at all, because of a paradigm-specific incompatibility instead of a formal incompatibility.

Another problem was noted in Section 3.4 about the occurrence of *-si-m*. Indeed, it does not surface when portmanteau *-mʔna* is present. I argued there that we can use Pāṇinian competition between *-mʔna-si* and *-si-m*, so that the former, more specific, will block the latter. This means expanding a bit the hierarchy we had for *-mʔna* in Figure 4.8. In addition to the divide in valence between transitive and intransitive cases, the number of morphs can now be either one or two. These two dimensions of variation intersect to give three leaf rules types: one monomorphic type for S scenarios, another monomorphic one for general A>3 scenarios, and the last, dimorphic one, specifically for A>3NSG scenarios. Pāṇinian competition will restrict the appearance of the second one so that it only becomes available in A>3SG scenarios. The bottom right rule in Figure 4.13 is now prelinked, with a dashed line, to the patient *-si* rule that is found in Figure 4.12: this is one way of capturing exceptional behavior. This will allow *-mʔna-si* to appear in a 1PE>3NSG.PRET function, thus preempting *-si-m*, which merely serves to express (1∨2)PL>NSG.

Finally, let me dwell upon a desirable property of the analysis, which resolves a case that was left aside in Section 4.3.2. Indeed, I ignored number expression in 1SG>2 scenarios then. This areas presents us with an intricate fact: expression of patient number is unavailable in 1>2, except when the agent is first-person singular. Therefore, this area is not fully uninflected. In Figure 4.12, notice that the *-ŋ* rule type can actually performs two functions: introducing the dependent *-ŋ*, on the one hand, and allowing

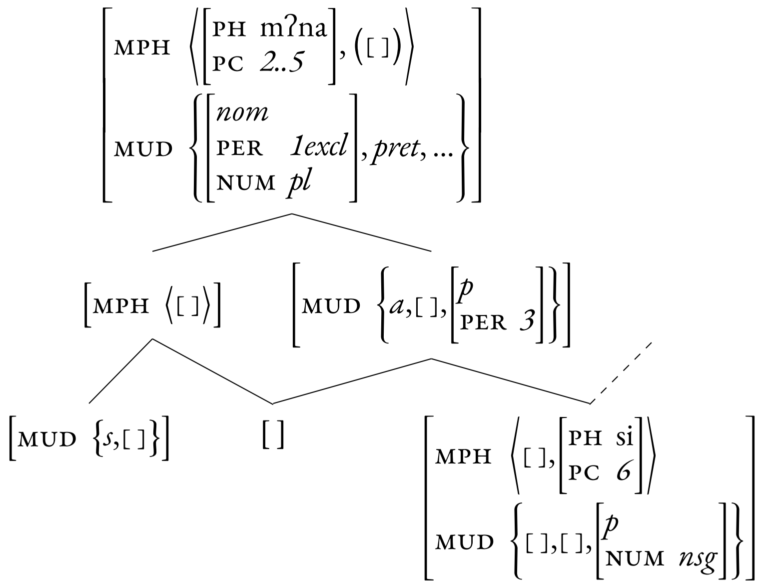

Figure 4.13: Rule hierarchy describing all rules involving *-mʔna*.

one last inflectedness condition for *-i* and *-si*. In this way, we can reinterpret the inflectedness in the 1SG>2 area as a form of reverse conditioning: it is the dependent that licenses the carrier, which would not otherwise have surfaced. This retrospectively justifies the integration of uninflectedness and dependent marking into one MORPHOTAX dimension.

4.5 Final analysis

While all plural markers from Figure 4.10 readily find their place in an expanded version of the hierarchy in Figure 4.12, the rules for *-mʔna* have not been integrated yet. To do this, we should make *-mʔna-si* into a subtype of the patient *-si* rule; then, it will need to be cross-classified with a type from the MORPHOTAX dimension. To reflect the fact that *-mʔna* does not take any dependent (or possibly one: *-si*), we can actually prelink its general supertype with MORPHOTAX in the hierarchy, thus obviating the need to match an independent leaf type in that dimension. Because *-mʔna* is already a plural marker, it can also be considered a subtype of the non-singular rule type. Therefore, *-mʔna*, as a very specific marker, naturally finds its place in the middle of the hierarchy. It is prelinked to both dimensions, thus reflecting its exceptional nature within the family of plural markers. The final version for the hierarchy is now presented in Figure 4.14;

the rules for *-mʔna* from Figure 4.13 are found within it as a subhierarchy.[11]

Among the markers that are not represented in the hierarchy are person markers: these do not pose any problem, however. They will simply be subtypes of the topmost *part-rr* supertype, that do not participate in uninflectedness or dependence phenomena like number markers do. Some of their own subtypes will be able to participate in this way, though: this is the case of *a-*, whose specific plural subtype is also distributed with respect to $\boxed{\text{MORPHOTAX}}$ to account for its distribution and competition with *-i*. Another generalization that we had noticed is the recurrent use of *-ŋ* in relation to first person singular, even outside dependent marking. Again, while this form–function association itself does not participate in the cross-classification, its subtype, dependent *-ŋ*, will. Neither of these other rule types are represented in Figure 4.14, but their presence is suggested by dashed links.

Let us take stock of our analytical journey. Section 4.2 exemplified the use of inheritance in typed feature structures to successfully reduce pseudo-Pāṇinian splits as a special case of Pāṇinian competition, which ultimately paved the way for a very similar analysis to explain the L-shaped split between *-si* and *mɛ-*. Then, we expanded our scope to plural markers, noticed that they were conditioned on specific areas of the paradigm, and accounted for this systematicity by means of a cross-classifying dimension to restrict their distribution. With minimal changes, this dimension could then be generalized to incorporate dependents next to those number markers which are carriers, and as a bonus, allowed us to settle the very last puzzle of number uninflectedness.

[11] As a final informational readjustment we did not get the chance to mention, we can capture the fact that *-m* also restricts inflection by moving the $\left[\text{MS} \ \left\{\left[\text{PER} \ \mathit{3}\right], \ldots\right\}\right]$ type up outside the monomorphic rule type in $\boxed{\text{MORPHOTAX}}$, and considering it a supertype of the *-m* dependent rule.

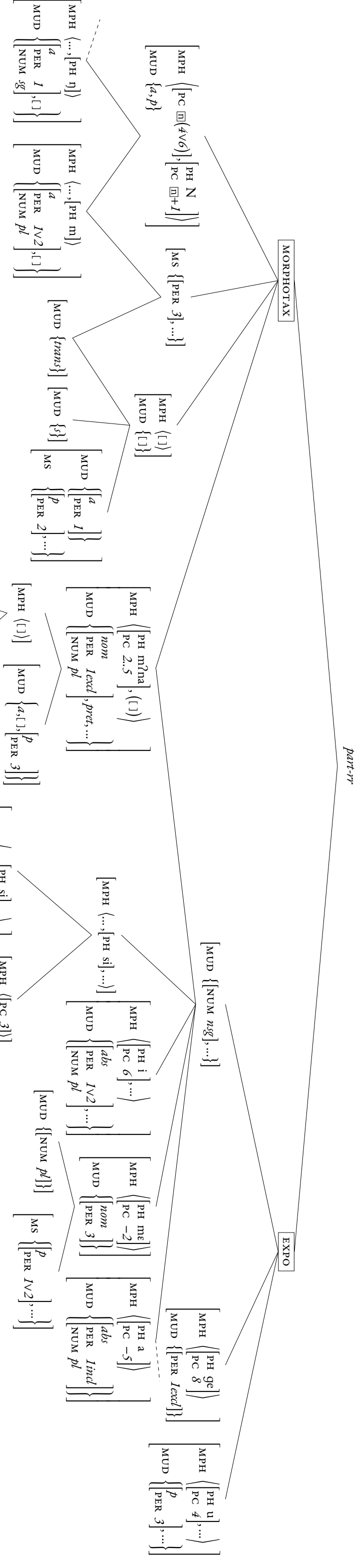

Figure 4.14: Final hierarchy of rules, accounting for all phenomena in Section 3.2.

Chapter 5

Conclusion

We have explored a few unusual patterns which arise in the Limbu conjugation system. For reasons of time and space, many aspects of the language's inflection had to be left out. For instance, negative marking is quite intricate as it can be conditioned on certain markers, and is also involved in copying phenomena (see Tables C.1). Or in terms of the person system, free pronouns, imperatives and more unusual constructions also probably structure speakers' representation of personal marking. A foretaste of the associated paradigms in Limbu and other languages is given in Appendix B.

Inflectional typology and morphological theory

Pseudo-Pāṇini This Master's dissertation brought out several inflectional problems in the Limbu system. Notably, unusual competition and uninflectedness gave rise to morphological splits that could be referred to as morphomic. I identified a subset of these as pseudo-Pāṇinian splits and showed how to use several mechanisms of IbM to successfully analyze these. Hopefully, the perspective that such a definition provides can help understanding superficially similar phenomena in other inflectional systems. This would be interesting both in a language-internal perspective, as for Limbu, and for wider morphological theory.

Morphomes and IbM Despite their intricate nature of the splits, a broad and systematic study of the system lay the ground for a fruitful analysis, accounting for all facts of competition and distribution in the system. I showcased the use of the formal mechanisms provided by IbM by solving the problems that had emerged. Thus, I was able to reduce the morphomic-like pattern of the splits to regularities and subregularities in the system by using the full range of analytic devices that this theory of morphology provides.

The power of IbM Within my analysis, a common theme that emerges is the ability of an inheritance logic of typed feature structures to capture the interplay of regularity and special conditioning in morphology. Specifically, we can encode partial regularities by separating dimensions of variation, which are then combined using dynamic cross-classification. On the other hand, exceptional behavior and conditions that escape our broad-stroke generalizations can always be accounted for using prelinking with different supertypes, or distinction of subcases, which are the ones that will be instantiated in actual word forms. Within this system, there emerges a tension between, on the one hand, the comprehension-oriented problem of identification and systematicity of form-to-function mapping, whose resolution is catalyzed by the macro-rule; and on the other, the production-oriented problem of finding the appropriate form to use (cf. the Paradigm Cell Filling Problem, Ackerman, Blevins & Malouf 2009), which for subregular patterns is taken care of by the micro-rules listing the class members, and Pāṇinian competition.

The devices that IbM uses to capture this tension have hopefully shown their power in dividing and conquering an especially intricate system; while being motivated at their core by simple principles of knowledge representation and knowledge use.

Morphomes and morphological theory The fact that a seasoned theory of morphology can account for near-morphomic patterns without recourse to ad-hoc lists of cells should raise a wider methodological issue, and encourage caution in evaluating certain types of splits. This applies even for gender systems of the Caucasus (cf. Table 2.2c), where some regularities and semantic principles are still found (Baerman, Brown & Corbett 2005: p. 86), as I mentioned in Section 2. It seems that there is a wide range of variation between landmark types of splits as those presented in Chapter 2, and Limbu markers like *-si* fall in the middle of this spectrum. Then, what are morphomes, exactly?

Pseudo-Pāṇini and diachrony The different use cases that are characteristic of pseudo-Pāṇinian splits give additional credit to the notion of morphological *bricolage*. While a speaker should be able to memorize the system, imperfect memory and general frequency facts may be quite powerful forces in the diachrony of inflection. Piecemeal analogy might be able to give rise to partially chaotic morphological change, and to preserve its results. Then, patterns like those of Limbu, which are not describable by a simple, closed featural description may well persist in a community, at least for some time. This is another argument in favor of underspecification in the representation of inflectional knowledge, and formalisms which embrace it like IbM.

Related phenomena in Kiranti languages

Within wider Kiranti, this work can find developments in several directions. Indeed, several of the inflectional themes developed here for Limbu seem to arise in other lan-

↓A \ P→	1SG	1DU	1PL
2SG	•	↑	↗
2DU	←	∅	→
2PL	↙	↓	↘

(a) Uninflectedness and override: Limbu

↓A \ P→	1SG	1DU	1PL
2SG	→	→	→
2DU	→	→	→
2PL	→	→	→

(b) Horizontal syncretism: South-East Camling

↓A \ P→	1SG	1DU	1PL
2SG	∅	↓	↓
2DU	→	→↓	→↓
2PL	→	→↓	→↓

(c) Expression of either number and overabundance: Athpare

↓A \ P→	1SG	1DU	1PL
2SG	∅	↓	P
2DU	→	→↓	→P
2PL	A	A↓	AP

(d) Cross syncretism: Thulung. *A* and *P* are two different markers expressing number.

Table 5.1: Patterns of syncretism in 2>1 scenarios in Kiranti languages

guages as well.

Other patterns of copying One such theme is that of copying: Camling and Athpare also find copying of *-m* and *-ŋ* in their system. Zimmermann (2016) has noted the wide range of such phenomena in Kiranti. For instance, copies in negative marking in Chhatthare Limbu (Stump 2017), reminiscent of optional copies in the Phedāppe Limbu of van Driem (1987). Athpare can copy certain markers to express tense. This abundance of copying phenomena offers exciting perspectives from a micro-typological perspective.

Other strategies of number marking One such theme is that of morphomic-like patterns of syncretism in other Kiranti languages, especially for number marking. For number marking, the 2>1 area of the paradigm is especially striking: this variation is represented in Table 5.1. Within a four-language sample, one can find four different strategies of marking, giving rise to four different patterns of syncretism (cf. Table B.4 for a fifth pattern in Khaling). This raises the question of how to analyze such a diversity. A possible lead might be to try to reuse the distinction between exponence and inflectedness from Section 4.3.2; changing the conditions for inflectedness while keeping strikingly similar exponents.

Other patterns of person syncretism Variation also affects person marking in Kiranti languages. The different patterns of syncretism are well expressed by van Driem

(1992: pp. 36–43)'s tables. It seems that closely related dialects like Chhatthare Limbu (Tumbahang 2007: pp. 32–37) and extensively-studied Bantawa (Cho 2020) provide a good starting point for micro-typological studies, to explore the different strategies for person marking. This could be studied in comparison with, for instance, Baerman, Brown & Corbett (2005)'s overview of person syncretism in transitive systems. It might be that functional explanations in the spirit of Heath (1991, 1998) and DeLancey (2018) emerge or are reinforced by such typological studies.

In sum, Kiranti languages seem to hide treasures of linguistic diversity by providing a family of inflectional systems with both great internal coherence and idiosyncrasies. Whether these idiosyncrasies can find functional motivations or not, the sheer space of variation that these languages map out can certainly offer insights or confirmation about the nature and implementation of inflectional morphology, via the structuring lens of formal analyses.

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Appendix A

Other hierarchies of rules

Figure A.1: Hierarchy accounting for specific exceptions to person marking (distribution of *-ʔɛ/aŋ/paŋ*). The more general association of /ŋ/ with first person singular is suggested by the incomplete dashed edge.

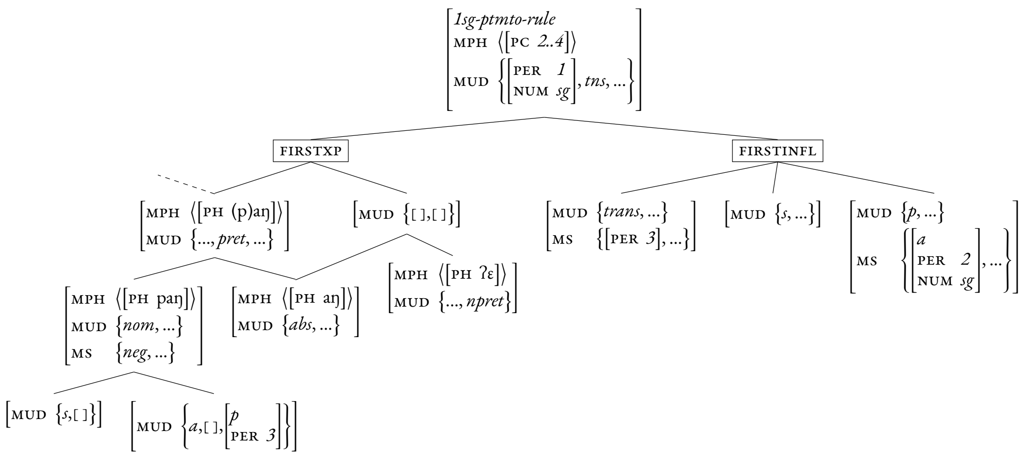

Figure A.2: Abstract rule type expressing association of /ŋ/ with first person singular

$$\begin{bmatrix} \text{MPH} & \left\langle \ldots, \begin{bmatrix} \text{PH} & \ldots\text{ŋ} \end{bmatrix} \right\rangle \\ \text{MUD} & \left\{ \begin{bmatrix} \text{PER} & \mathit{1} \\ \text{NUM} & \mathit{sg} \end{bmatrix}, \ldots \right\} \end{bmatrix}$$

Appendix B

Other verbal TAM configurations

Apart from assertions inflecting for two participants, orders and commands still usually require marking A number, and P person–number if relevant. The morphological material for these seems to be less discussed in descriptive work.

Adhortative

Limbu marks adhortative, i.e. 'let's' forms (where A or S is a first-inclusive participant), by removing the *a-* prefix from the 1INCL non-preterite forms, in accordance with their referential and temporal function (van Driem 1987: p. 184). Van Driem analyzes *a-* as a general first-person marker; I prefer an analysis of first-person, conditioned on some second person being relevant in the scenario ("1∧2": cf. p. 53f and Figure 4.10). Just like, cross-linguistically, second-person marking can be obviated in imperatives, we can construe the absence of *a-* there as a general phenomenon whereby commands may not mark the person they are directed at. Interestingly, though plural marking is recovered in intransitive scenarios. Indeed, recall from Section 4.3.1 that in non-adhortative "simplex" verbs, *a-* suffices as a marker of (3>)1INCL and preempts the usual 1∨2PLP/S number marker *-i*. Exceptionally in the adhortative, when *a-* is unavailable, *-i* surfaces again (bolded cell in Table B.1a). Assuming plural *a-* is understood as a subtype of general *a-* 1∧2 as in Figure 4.10, we can then analyze *-i* as a general 1∨2PL marker which is preempted not in all 1INCL scenarios, but specifically in *non-adhortative* 1INCL scenarios.

A so-called "nexal" negator suffix *-me:n* is used, instead of the usual consisting of a prefix and a suffix; it can be used in general to negate any clause and can be translated as 'it is not the case that' (van Driem 1987: p. 60). Naturally, in negations, it has a prohibitive meaning.

Imperative

Limbu imperatives involve a specific final *-ɛʔ* suffix and no personal prefix, along with conditioned number marking: *-ɛʔ* 2SG, *-ɛtch-ɛʔ* 2DU, *-amm-ɛʔ* 2PL. Negation is marked

Table B.1: Limbu adhortatives

(a) Adhortative

↓ A \ P →	3SG	3DU	3PL	↓ S
1DI	-s-u	-s-u-si	→	-si
1PI	-u-m	-u-m-si-m	→	**-i**

(b) Indicative non-preterite first-person-inclusive forms

↓ A \ P →	3SG	3DU	3PL	↓ S
1DI	a- -s-u	a- -s-u-si	→	a- -si
1PI	a- -u-m	a- -u-m-si-m	→	a-

with a fuller prefix than in non-imperatives: *mɛn-*. In transitive scenarios, third-person P is unmarked. Non-singular third-person P, however, does have an *-ɛs* suffix coming between second-person number and imperative marking. Interestingly, haplology means that only one marker of number surfaces in 2DU>3DU.IMP.AFF scenarios (in bold in Table B.2a), a pattern reminiscent from all Thulung DU>DU combinations (Lahaussois 2020: p. 96), and Camling DU>3NSG scenarios.

In keeping with non-imperative forms, *a-* alone (without *kɛ-*), can be used to signal first-person P in imperatives: that is, in 2>1 scenarios. Just like the corresponding non-imperative form, number is not differentiated in 2>1.IMP scenarios (van Driem 1987: p. 190). Unusual is the fact that negated imperatives with *a-* are not permitted and instead require a free word *na:pmi*, generally meaning 'someone else' but sometimes signaling first-person P in 2>1 scenarios (van Driem 1987: p. 477). This strategy is also common outside the imperative and outside the negative (van Driem 1987: p. 78). The negated version of *-aŋ-ɛʔ* 2SG>1SG.IMP is attested, however (van Driem 1987: p. 191).

Table B.2: Limbu imperatives

(a) Affirmative imperative

↓A \ P→	1SG	1DE	1PE	3SG	3DU	3PL	↓S
2SG	-aŋ-ɛʔ	↑	↗	-ɛʔ	-ɛs-ɛʔ	→	-ɛʔ
2DU	←	a- -ɛʔ	→	-ɛtch-ɛʔ	**-ɛs-ɛʔ**	→	-ɛtch-ɛʔ
2PL	↙	↓	↘	-amm-ɛʔ	-am-s-ɛʔ[a]	→	-amm-ɛʔ

[a] A fuller, copied form is attested, but rarer: *-am-si-m-ɛʔ* (van Driem 1987: p. 192). This suggests the analysis can be unified with the non-imperative, though special care is needed to account for the *-ɛs* allomorph of *-si* in 2SG/DU>3NSG forms.

(b) Negative imperative

↓A \ P→	1SG	1DE	1PE	3SG	3DU	3PL	↓S
2SG	mɛn- -aŋ-ɛʔ	↑	↗	**mɛn- -ʔɛʔ**	mɛn- -ɛs-ɛʔ	→	**mɛn- -ʔɛʔ**
2DU	←	na:pmi mɛn- -ɛʔ	→	mɛn- -s-ɛʔ	mɛn- -s-ɛs-ɛʔ	→	mɛn- -s-ɛʔ
2PL	↙	↓	↘	mɛn- -amm-ɛʔ	mɛn- -am-s-ɛʔ,	→	mɛn- -amm-ɛʔ

Table B.3: Thulung imperatives (Lahaussois 2020: p. 109)

↓A \ P→	1SG	1DE	1PE	3SG	3DU	3PL	↓S
2SG	?	?	?	-[d]a	-[d]at͡siʔ	-[d]amiʔ	-[d]a
2DU	?	?	?	-t͡si	-t͡si	-t͡simi	-t͡si
2PL	?	?	?	-ni	-nit͡si	-nimi	-ni

Table B.4: Khaling imperatives (Jacques et al. 2012: pp. 1114, 1118)

↓A \ P→	1SG	1DE	1PE	3SG	3DU	3PL	↓S
2SG	-ʌje	-uje	-kʌje	-e	↑	↑	-e
2DU	-ʌsuje	↓	↓	←	-ije	↑	-ije
2PL	-ʌnuje	↓	↓	←	←	-nuje	-nuje

Table B.5: Athpare imperatives (Ebert 1997a: p. 65). Superscript -^{a} marks the so-called "past base", which includes an /a/ segment, used notably in certain past forms with a consonant-initial suffix.

(a) Affirmative

↓A \ P→	1SG	1DE	1PE	3SG	3DU	3PL	↓S
2SG	-^{a}ŋ	?	?	-u	?	?	-^{a}
2DU	?	?	?	?	?	?	?
2PL	?	?	?	-^{a}n-u-m	?	?	-^{a}ni

(b) Negative

↓A \ P→	1SG	1DE	1PE	3SG	3DU	3PL	↓S
2SG	-ni-ŋ	?	?	-u-n	?	?	-nen, -ni
2DU	?	?	?	?	?	?	?
2PL	?	?	?	-^{a}n-u-m	?	?	-^{a}ni

Table B.6: Camling imperatives (Ebert 1997b: 32f)

↓A \ P→	1SG	1DE	1PE	3SG	3DU	3PL	↓S
2SG	-unga	?	?	-u	?	?	∅
2DU	?	?	?	?	?	?	-ci
2PL	?	?	?	?	?	?	-ni

Appendix C

Full inflectional tables for Limbu

[1] In his lists, van Driem (1987: p. 372) sometimes uses *ke-*, expressing second person, instead of the usual *kɛ-*. More precisely, this happens in some but not all affirmative forms which happen not to have a following prefix (*mɛ-* or *n-*). In the corresponding negative, they become *kɛn-*. The distribution of this variant seems to form a morphological natural class: 3SG>2.AFF. This does not seem to be mentioned in chapter 4 where verbs are analyzed, and only occurs in the list for the verb *huʔmaʔ* (pp. 368–374). I take this to be a mistake and do not try to account for it, keeping *kɛ-* everywhere.

Table C.1: Concrete table for Limbu transitive person marking: all basic tense–polarity–person combinations are shown. Inflection recovered from conjugation lists in van Driem 1987: pp. 368–374. All errors are mine only.

Polarity: AFF

Tense	AFF.NPRET	1SG	1DE	1PE	1DI	1PI	2SG	2DU	2PL	3SG	3DU/PL
	1SG						- -nε	- -nεtchiŋ	- -niŋ	- -uŋ	- -uŋsiŋ
	1DE						←	- -nεtchige	→	- -suge	- -susige
	1PE						↙	↓	↘	- -umbe	- -umsimbe
	1DI									a- -su	a- -susi
NPRET	1PI									a- -um	a- -umsim
	2SG	kε- -ʔε	↑	↗						kε- -u	kε- -usi
	2DU	←	agε- -	→						kε- -su	kε- -susi
	2PL	↙	↓	↘						kε- -um	kε- -umsim
	3SG	- -ʔε	- -sige	- -ige	a- -si	a- -	kε- -[1]	kε- -si	kε- -i	- -u	- -usi
	3DU	↑	↑	↑	↑	↑	↑	↑	↑	- -su	- -susi
	3PL	mε- -ʔε	mε- -sige	mε- -ige	am- -si	am- -	kεm- -	kεm- -si	kεm- -i	mε- -u	mε- -usi

Tense	AFF.PRET	1SG	1DE	1PE	1DI	1PI	2SG	2DU	2PL	3SG	3DU/PL
	1SG						- -nε	- -nεtchiŋ	- -niŋ	- -uŋ	- -uŋsiŋ
	1DE						←	- -nεtchige	→	- -εtchuge	- -εtchusige
	1PE						↙	↓	↘	- -mʔna	- -mʔnasi
	1DI									a- -εtchu	a- -εtchusi
PRET	1PI									a- -um	a- -umsim
	2SG	kε- -aŋ	↑	↗						kε- -u	kε- -usi
	2DU	←	agε- -ε	→						kε- -εtchu	kε- -εtchusi
	2PL	↙	↓	↘						kε- -um	kε- -umsim
	3SG	- -aŋ	- -εtchige	- -ige	a- -εtchi	a- -ε	kε- -ε	kε- -εtchi	kε- -i	- -u	- -usi
	3DU	↑	↑	↑	↑	↑	↑	↑	↑	- -εtchu	- -εtchusi
	3PL	mε- -aŋ	mε- -εtchige	mε- -ige	am- -εtchi	am- -ε	kεm- -ε	kεm- -εtchi	kεm- -i	mε- -u	mε- -usi

Polarity: NEG

Tense	NEG.NPRET	1SG	1DE	1PE	1DI	1PI	2SG	2DU	2PL	3SG	3DU/PL
	1SG						mε- -nεn	mε- -nεtchiŋnεn	mε- -niŋnεn	mε- -ʔεn	mε- -ʔεntchin
	1DE						←	mε- -nεtchigen	→	mε- -sugen	mε- -susigen
	1PE						↙	↓	↘	mε- -umben	mε- -umsimben
	1DI									an- -sun	an- -su(n)sin
NPRET	1PI									an- -umnεn	an- -umsimnεn
	2SG	kεn- -ʔεn	↑	↗						kεn- -un	kεn- -u(n)sin
	2DU	←	agεn- -nεn	→						kεn- -sun	kεn- -su(n)sin
	2PL	↙	↓	↘						kεn- -umnεn	kεn- -umsimnεn
	3SG	mε- -ʔεn	mε- -sigen	mε- -igen	an- -sin	an- -nεn	kεn- -nεn	kεn- -sin	kεn- -in	mε- -un	mε- -u(n)sin
	3DU	↑	↑	↑	↑	↑	↑	↑	↑	mε- -sun	mε- -su(n)sin
	3PL	mεn- -ʔεn	mεn- -sigen	mεn- -igen	amεn- -sin	amεn- -nεn	kεmεn- -nεn	kεmεn- -sin	kεmεn- -in	mεn- -un	mεn- -u(n)sin

Tense	NEG.PRET	1SG	1DE	1PE	1DI	1PI	2SG	2DU	2PL	3SG	3DU/PL
	1SG						mε- -nεn	mε- -nεtchiŋnεn	mε- -niŋnεn	mεn- -paŋ	mεn- -paŋsiŋ
	1DE						←	mε- -nεtchigen	→	mε- -εtchugen	mε- -εtchusigen
	1PE						↙	↓	↘	mεn- -mʔna	mεn- -mʔnasi
	1DI									an- -εtchun	an- -εtchu(n)sin
PRET	1PI									an- -umnεn	an- -umsimnεn
	2SG	kεn- -aŋnεn	↑	↗						kεn- -un	kεn- -u(n)sin
	2DU	←	agεn- -εn	→						kεn- -εtchun	kεn- -εtchu(n)sin
	2PL	↙	↓	↘						kεn- -umnεn	kεn- -umsimnεn
	3SG	mε- -aŋnεn	mε- -εtchigen	mε- -igen	an- -εtchin	an- -εn	kεn- -εn	kεn- -εtchin	kεn- -in	mε- -un	mε- -u(n)sin
	3DU	↑	↑	↑	↑	↑	↑	↑	↑	mε- -εtchun	mε- -εtchu(n)sin
	3PL	mεn- -aŋnεn	mεn- -εtchigen	mεn- -igen	amεn- -εtchin	amεn- -εn	kεmεn- -εn	kεmεn- -εtchin	kεmεn- -in	mεn- -un	mεn- -u(n)sin

Appendix D

Detailed person ontology

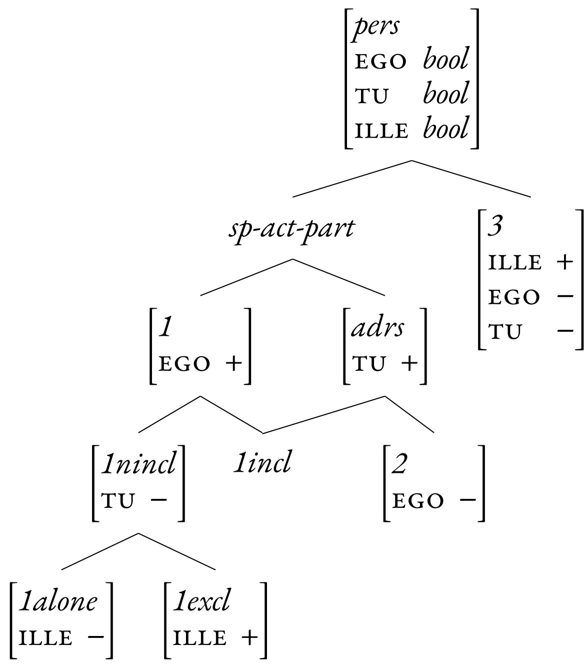

Figure D.1: Type hierarchy of person values with features inspired by those of Silverstein (1986).